Skydroid C14PRO 用户手册

小型高清四光三轴高精度云台

V1.1

建议：在您阅读本用户手册时，边阅读边操作。您在阅读这些说明时，如遇到困难请查阅**本用户手册**或致电我们售后（400-6996-520）及云卓微信公众平台，云卓官方群：318480806查看相关问题问答。

云卓官方群2群　　　　云卓微信公众平台

售后服务条款

1.本条款仅适用于云卓科技（福建）有限公司所生产的产品，云卓授权经销商销售的产品亦适用本 条款。
2.本公司产品自购买之日起，一周内经我司核实为非人为造成的质量问题，由云卓承担返修产 品的往返快递费，购买云卓产品一周以上一年以内经我司核实为质量问题，用户和云卓各自承 担寄出返修产品的快递费。
3.返修时需提供购买凭证和保修卡或网络平台交易记录。
4.云卓产品自购买之日起七天内，在正常使用情况下出现非人为造成的质量问题，外观无损坏， 凭保修卡及购机凭证在经销商处协商可以免费更换同型号产品；经销商在收到更换产品时烦请 第一时间通知云卓公司予以备案更换。
5.云卓产品将由云卓科技（福建）有限公司提供终身售后服务， 属于非人为造成的质量问题，一年内免费保修；对于自购买之日起人为的损坏、改装、拆机及超过一年免费保修期的，用户须支付 往返邮费及维修费用。
6.为确保您的权益受到保护，并能及时有效的为您服务，请在购买云卓产品时完整填写好保修 卡及索要购机凭证。用户享受本售后服务条款必须提供保修卡及购机凭证。
7.返修产品将于云卓公司收到后的 15 个工作日内寄回给顾客，并附上维修报告。
8.以上售后服务条款仅限于中国大陆销售的云卓产品。
港澳台及海外客户的售后问题发至邮箱sales01@skydroid.xin，具体售后细则视情况而定。

目录

警告和免责申明

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C14PRO云台出厂前已根据其所搭载的相机和镜头完成调试，切勿自行调整云台或者改变其机械机构，也不要为云台增加其他外接设备。云台结构精密，请勿自行对C14PRO作任何拆装，否则将会 导致云台相机工作异常。

为了您和他人的安全，请确保云台调试时，飞行器处于安全的状态，不在易燃易爆及有儿童的 环境下通电调试，强烈建议您在调试前取下飞行器的螺旋桨。

A）环境温度： -10°C ~ +60°C

B）贮存温度： -20°C ~ +60°C

C）大气压力： 86kPa~106kPa

D）使用地点不允许有爆炸危险的介质，周围介质中不应含有腐蚀金属和破坏绝缘的气体及导 电介质 ,不允许充满水蒸气及有严重的霉菌存在。

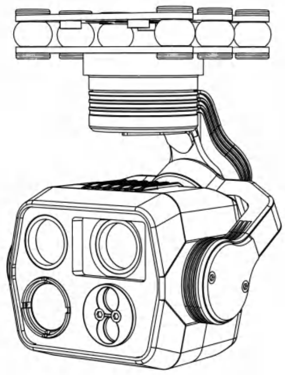
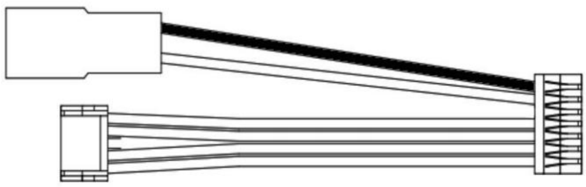
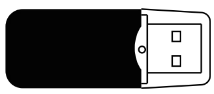
E）使用地点应具有防御雨、雪、风、沙、灰的设施。

由于我公司无法控制用户的具体使用、安装、改装以及使用不当等情况。由以上所造成的直接、 间接损失或损伤，我公司将不承担相应的损失及赔偿责任。如使用、安装、组装云卓科技产品，相应的结果由用户承担。因使用本产品而造成的间接或直接损失与伤害，我公司概不负责。

一、产品简介

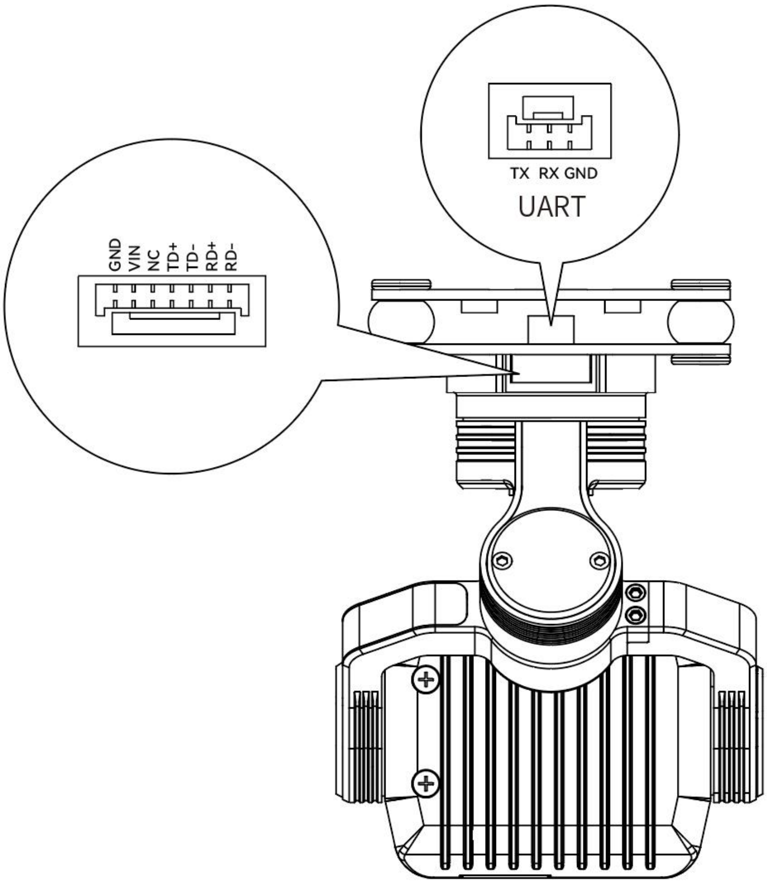
C14PRO是一款三轴高精度四光云台，专为无人机平台打造，集可见光、热成像与激光测距功能于一体，满足多场景作业需求。它搭载48MP高清可见光相机，支持100倍混合变焦，短焦与长焦摄镜头可自动灵活切换，搭配640x512分辨率热成像相机，探测距离达1.1km，实现昼夜全天候感知。集成1200m激光测距模块，精度最高±0.5m，大幅提升目标定位效率。三轴机械高精度系统将悬停抖动控制在±0.01°、飞行抖动±0.02°，输出稳定流畅画面;支持RTSP视频流传输与UDP控制协议，适配无人机网口接入。15V~80V宽压供电、-10°C~+50°C宽温工作，搭配吊装/倒装模式与全局回中功能，210g轻量化机身适配多型无人机平台，广泛应用于巡检、搜救、安防等场景。

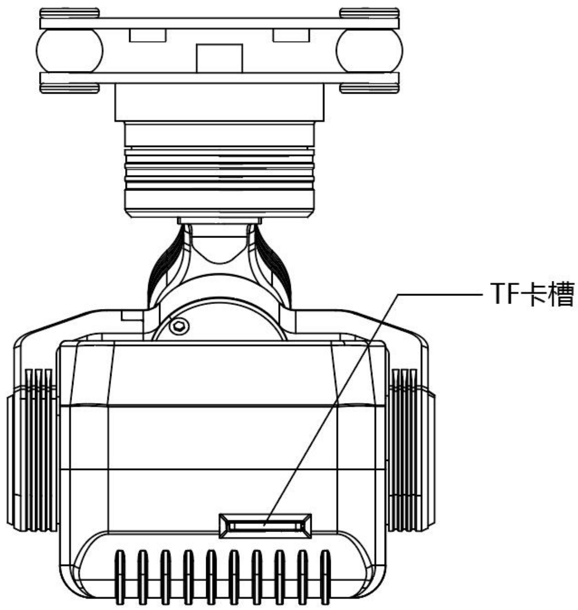
二、产品清单

C14PRO主体×1	
云台固定螺丝(M2.5*6) ×4	
网口电源线×1	
内存卡×1 (已安装)	
读卡器×1	

三、产品概览

接口及定义说明





四、产品参数

C14PRO整体参数	
视频输出信号接口	网口
控制信号输入接口	网口
工作电压	15V~80V
工作电流	15V376mA
工作温度	-10℃~+50℃
存储温度	-20℃~+50℃
重量	210g
外形尺寸	65.5mm(长)*68.57mm(宽)*92mm(高)
可控角度范围	-110°~+75°(俯仰)；-165°~+165°(水平);-40°~+40°(横滚)
增稳方式	三轴机械增稳
角度抖动量	悬停：±0.01° 飞行：±0.02°
视频输出	RTSP
控制方式	UDP
吊装/倒装模式	支持
全局回中/一键向下/偏航回中	支持

C14PRO可见光相机参数			
镜头像素	48MP	卡录视频分辨率	3840*2160
图像传感器	1/2.0	视频存储格式	H265
图传分辨率	1280*720	拍照分辨率	8000*6000
支持文件系统	FAT32/exfat	照片存储格式	JPG
混合变焦	100倍	支持存储卡类型	支持MicroSD存储(最大128G)

C14PRO短焦镜头参数			
光圈	F=2.6±5%	焦距	5.4mm±5%
HFOV/VFOV/DFOV	61.4°/47.9°/73.8°		

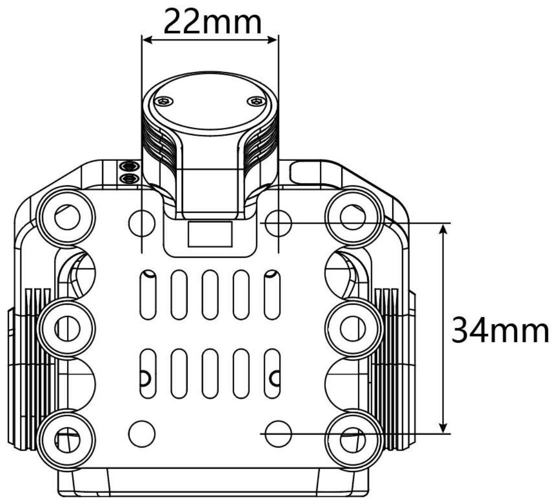
C14PRO长焦镜头参数			
光圈	F=2.4±10%	焦距	25mm±5%
HFOV/VFOV/DFOV	14.7°/11°/18.3°		

C14PRO热成像相机参数			
分辨率	640*512	光圈	F=1.0
焦距	11mm	像元间距	10μm
帧频	30Hz	响应波段	8~14μm
热时间常数	< 12ms	视场角	32.84°x26.35°
探测距离	1.1km		

C14PRO激光测距参数			
工作波段	905nm±5nm	探测范围	0.1M~1200M
测距能力	0.1m-1200m max1500m(夜间)【太阳大或能见度低则测程受一定影响】	测距精度	±0.5m (≤100m) ±0.5m+d*0.7‰ (100m<d<1200m)
接收口径	Φ6.5mm	测距频率	1.5~10Hz
虚警率	≤1%	准确率	≥98%

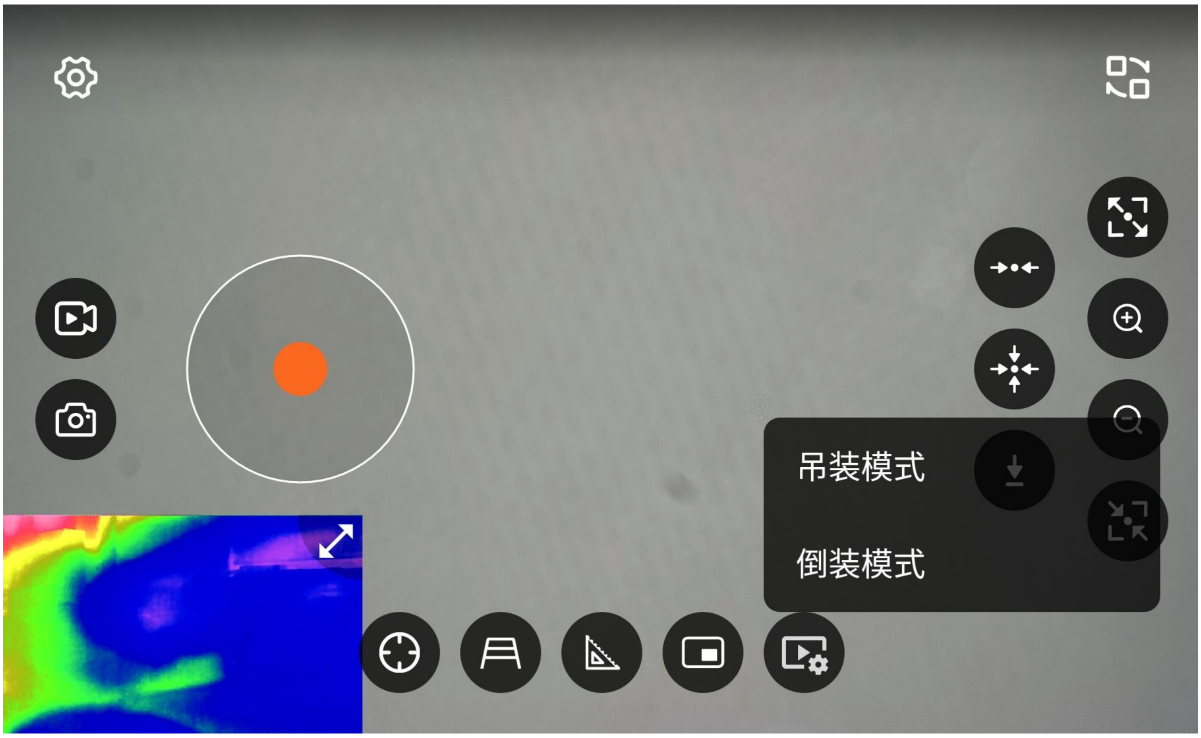
五、安装与调试

5.1 螺丝孔位与间距示意图如下：



注：用于云台的螺丝规格为 M2.5*6，用量4。
云台不使用时请勿悬挂于飞行器上，长期悬挂会加速避震球变形导致避震效果下降出现果冻现象。
云台安装时减震板之间必须保持相互间的绝对垂直与平行，不正确的安装将引起避震球变形导 致避震效果下降和无法自检。

5.2 云台工作模式

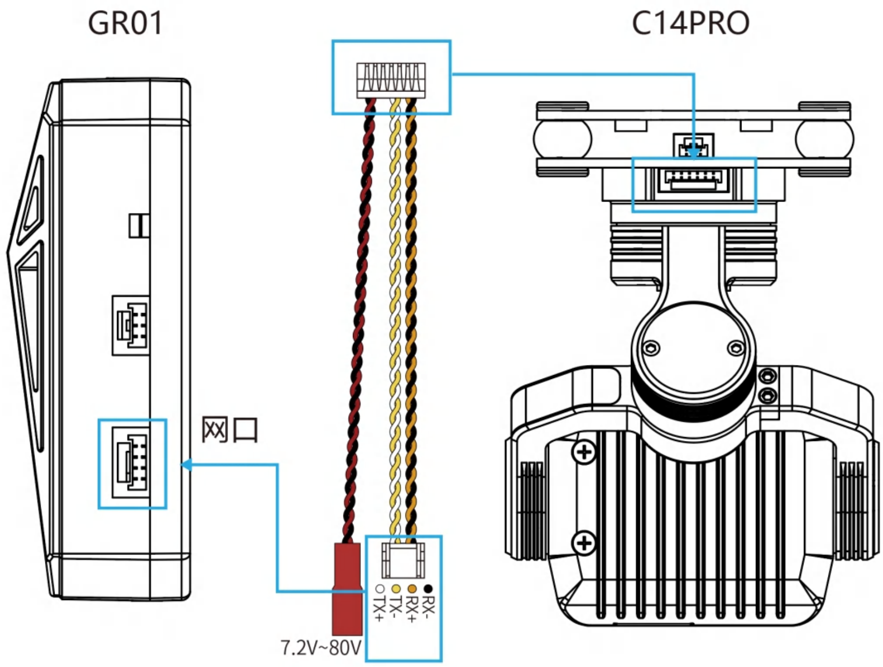


可在云台FPV设置吊装模式或者倒装模式

注意：出场默认吊装模式，请勿倒装上电；

设置好工作模式后，云台将会自动重启，图像自动进行翻转，必须将云台进行翻转安装为设置对应的模式放置，错误的放置方式将会导致云台电机自检不过导致损坏。

5.3 云台线路连接图及说明



①电源	红色JST-2P公头，供电电源:15V-80V (直流电源或者4S-18S锂电池)	
②网口线 (信号传输)	网络IP信号	RX-：网络IP信号 RX+：网络IP信号 TX-：网络IP信号 TX+：网络IP信号
视频传输	可见光视频输出 RTSP码流rtsp://192.168.144.108:554/stream=1	
	热成像视频输出 RTSP码流rtsp://192.168.144.108:555/stream=2	

5.4 C14PRO的使用

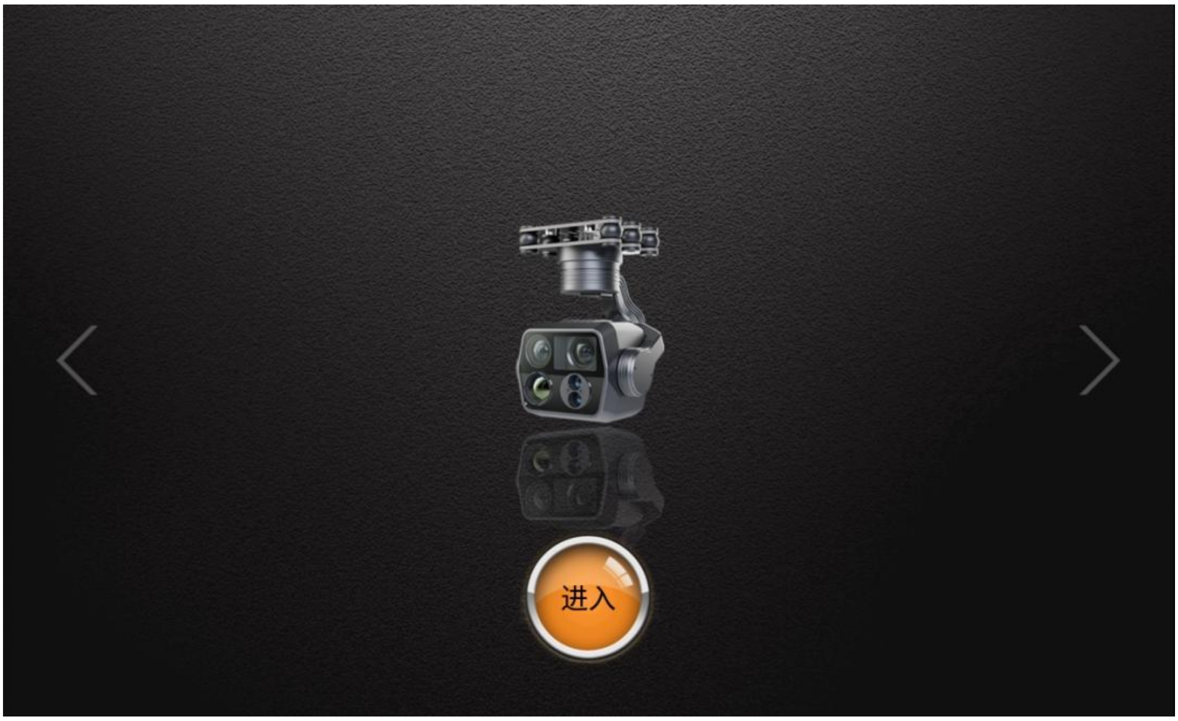
遥控器安装最新版云台FPV软件，并打开。

软件地址：http://file.skydroid.xin/SkydroidCameraFPV.apk 

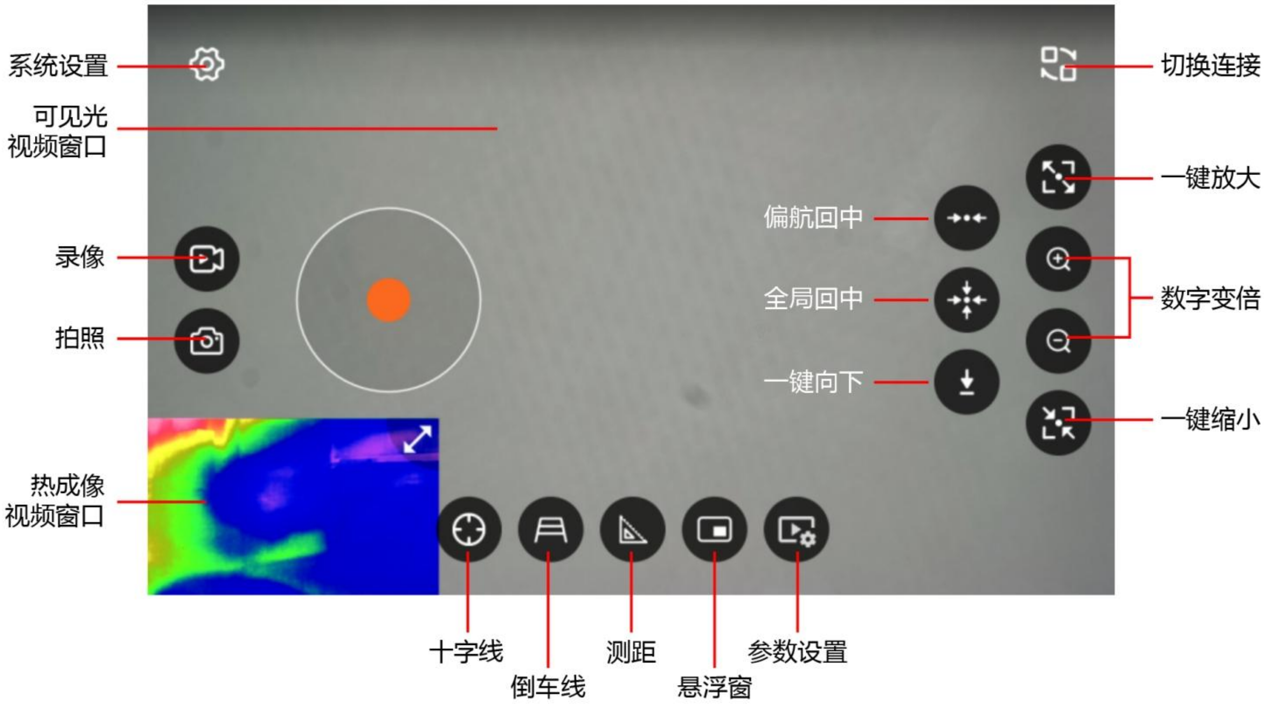
①云台启动

云台安装固定好后，供电，云台会启动，请等待启动完成。

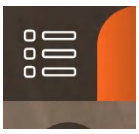
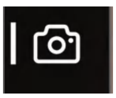
② 打开云台FPV，选择C14PRO图案的连接方式，点击进入。



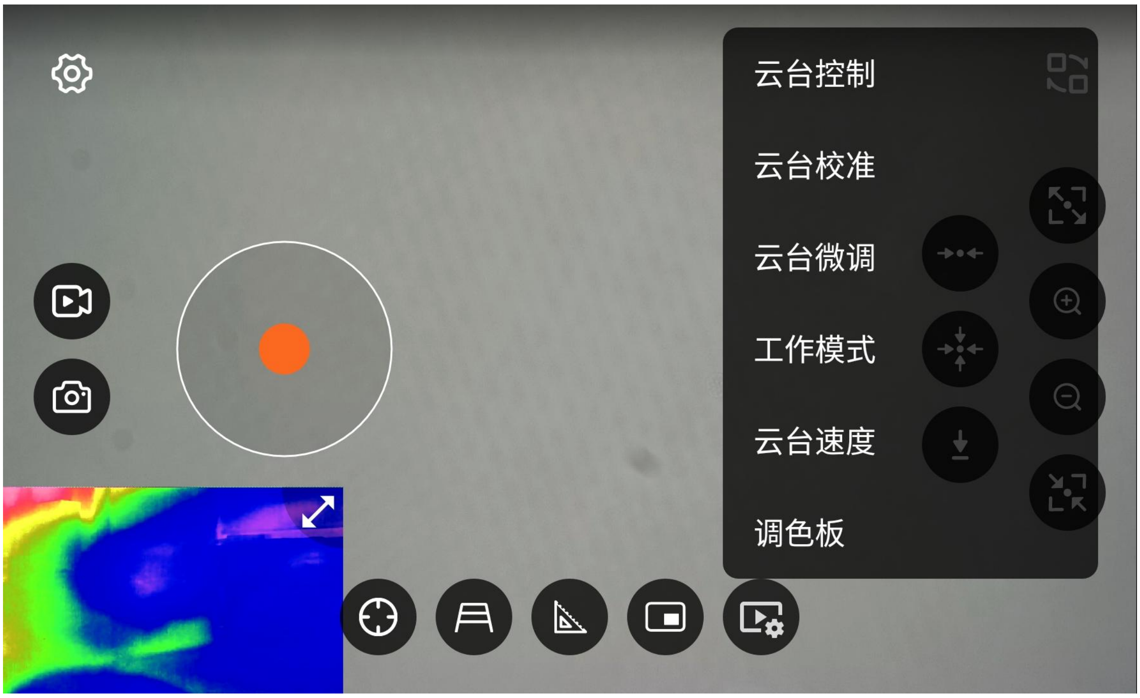
③云台FPV主页介绍



④FLY GCS画面设置 云卓 Skydroid FLY下载：http://file.skydroid.xin/SkydroidFly.apk

APP首页左上角  ->  ->左窗口点开，设置为C14/C14PRO

5.5 云台的参数设置

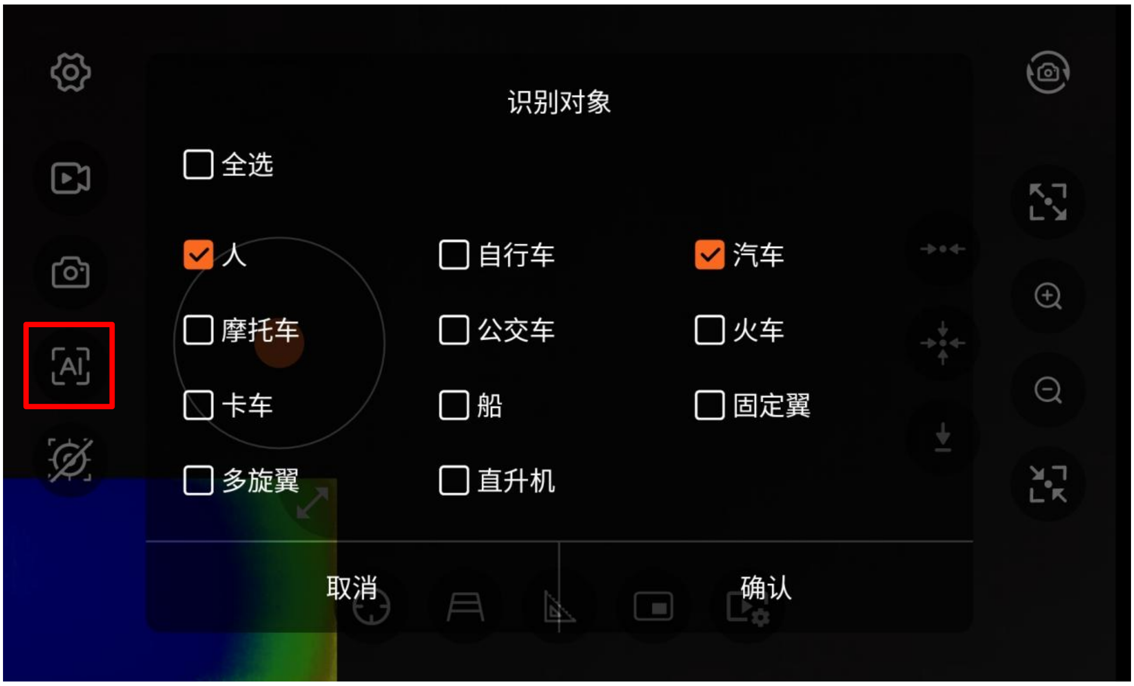


参数设置里可以设置云台控制、校准、微调、工作模式、速度、调色板等功能。

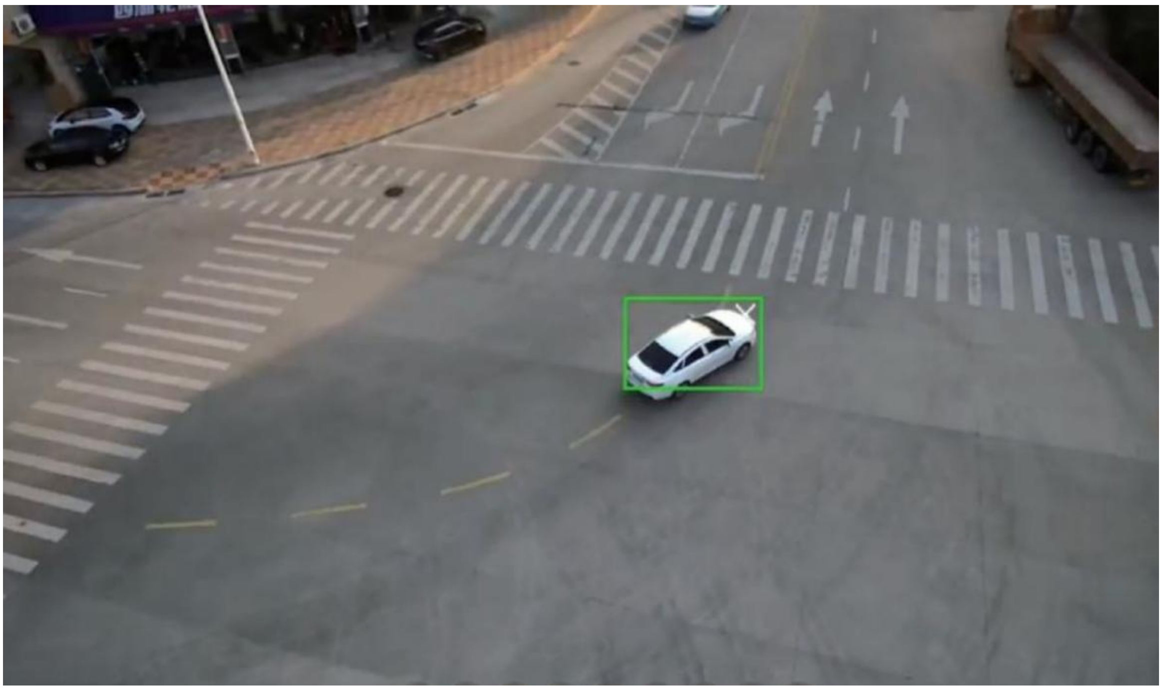
5.5.1 云台控制

点击参数设置->云台控制，有Ai跟随和三种控制方式可供选择，可同时勾选多种方式 （Ai跟随不能和手势控制同时使用）

①云台Ai目标跟随，点击AI，可单选、多选或全选需要自动识别跟随的物体。

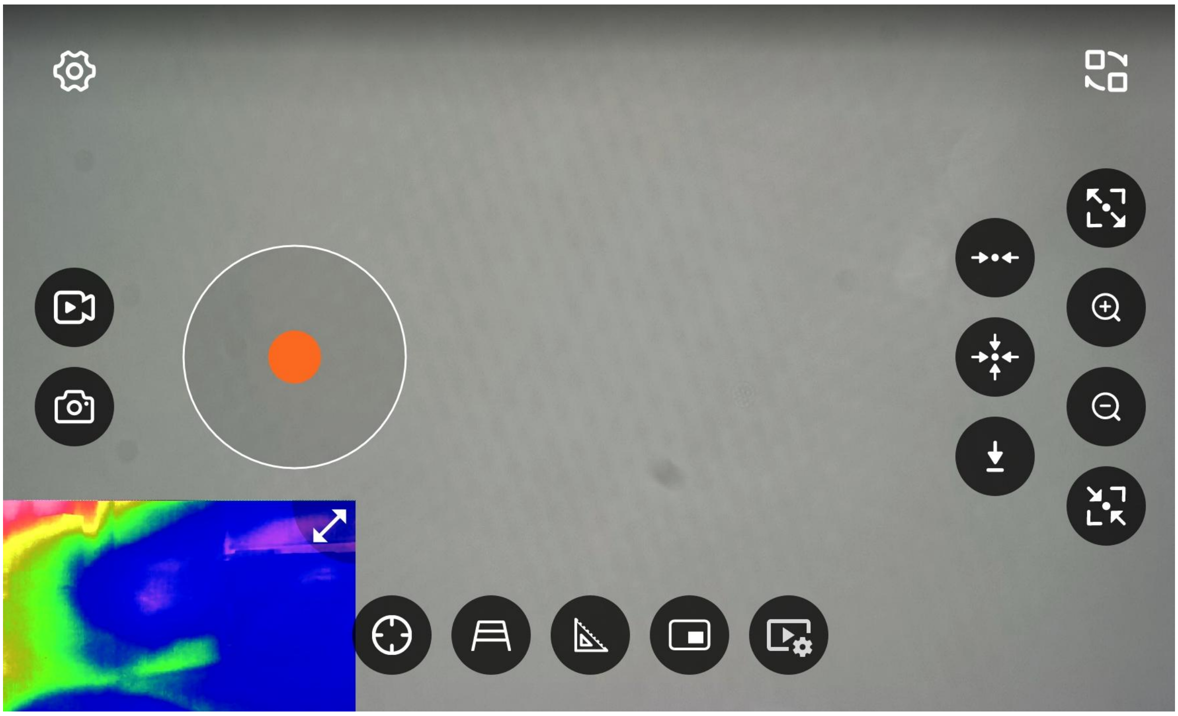


C14PRO将自动识别物体（绿框为已识别）

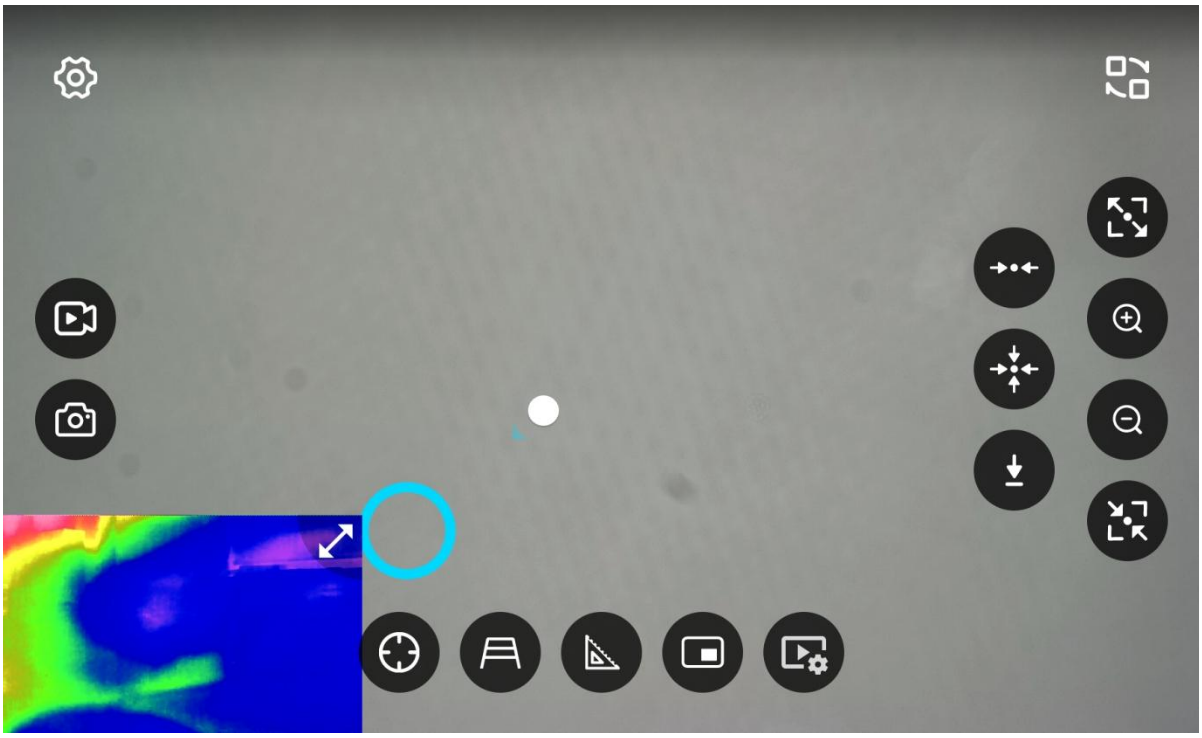


点击框内需要跟随的物体，将变为白色框（白框为已锁定），
C14PRO将自动追踪目标

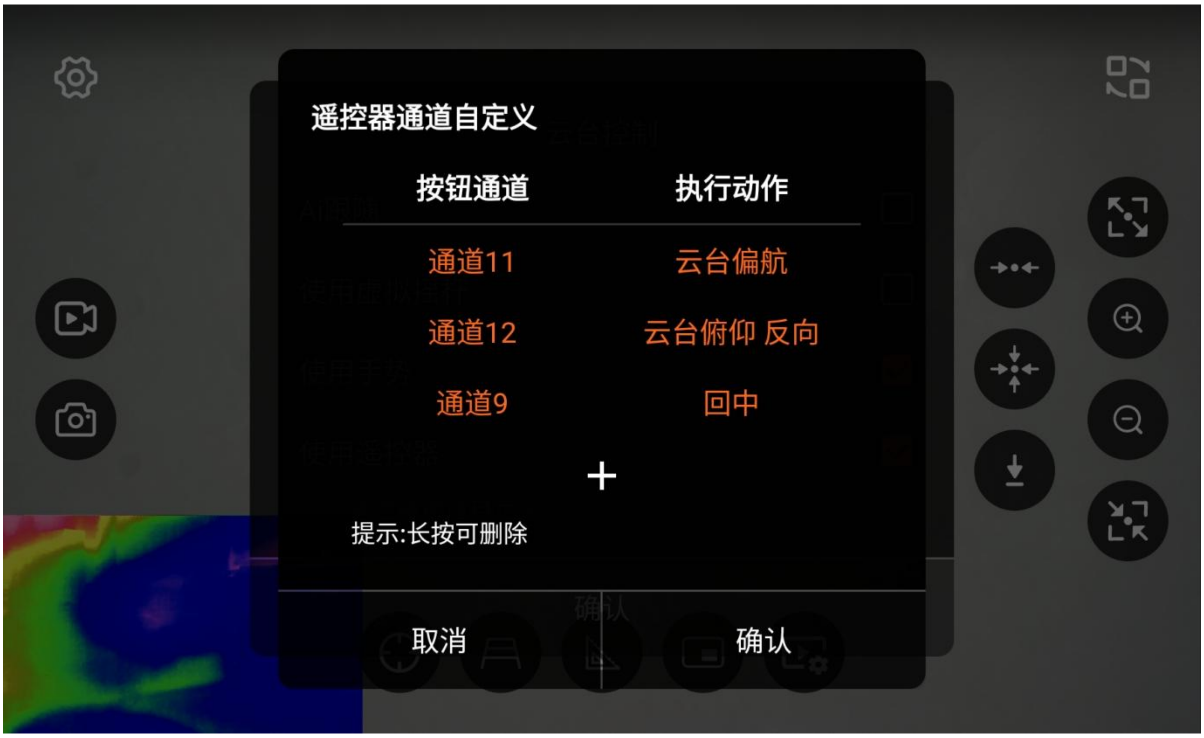
②虚拟摇杆控制，可通过虚拟摇杆控制云台转向和俯仰，且支持一键回中等功能。



③手势控制，通过滑动屏幕控制云台转向和俯仰。



④遥控器通道控制,通过自定义遥控器通道控制云台转向和俯仰，以及拍照录像等功能（遥控 器通道可通过遥控器助手->舵量查看查询）



5.5.2 云台校准

温度校准：校准云台工作时的温度，以免因环境温度与IMU工作温度差异过大导致云台无法正常工作。温度校准出厂已经校准，若无必要，请勿自己校准，校准前请联系云卓相关 技术人员确认获取校准密码。

5.5.3 校准锅盖效

注：锅盖效应出厂已经校准，因校准方式特殊，若无必要，请勿自己校准，校准前请联系云卓相关 技术人员确认获取校准密码。

校准方法:

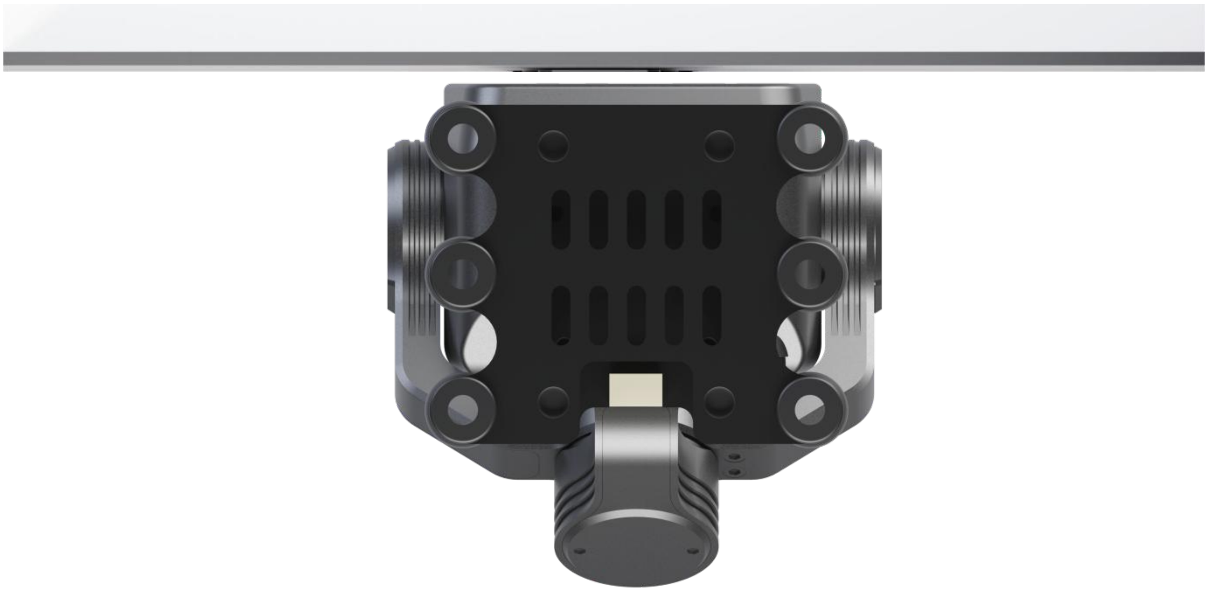
①C14PRO通电预热五分钟以上。

②选择一块表面平整、温度均匀、磨砂均匀、反光率低的面板。

③移动C14PRO靠近接触靶面，靶面的成像覆盖全部视场无杂散光进入为最佳。

④点击校准锅盖效应，画面为均匀灰色后校准成功。

锅盖标定靶面示意图

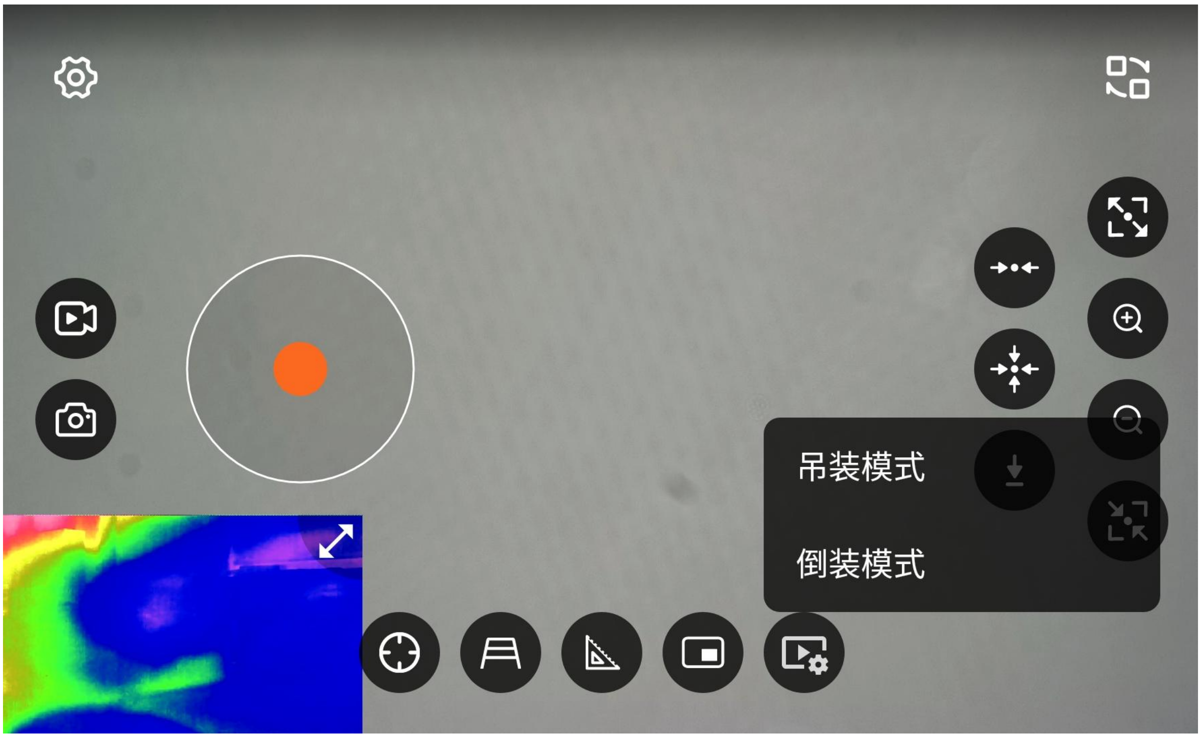


校准锅盖效应示意图

云台微调：微调云台的水平轴与俯仰轴。

注：出厂已做好云台校准，若无出现云台问题，请勿点击。

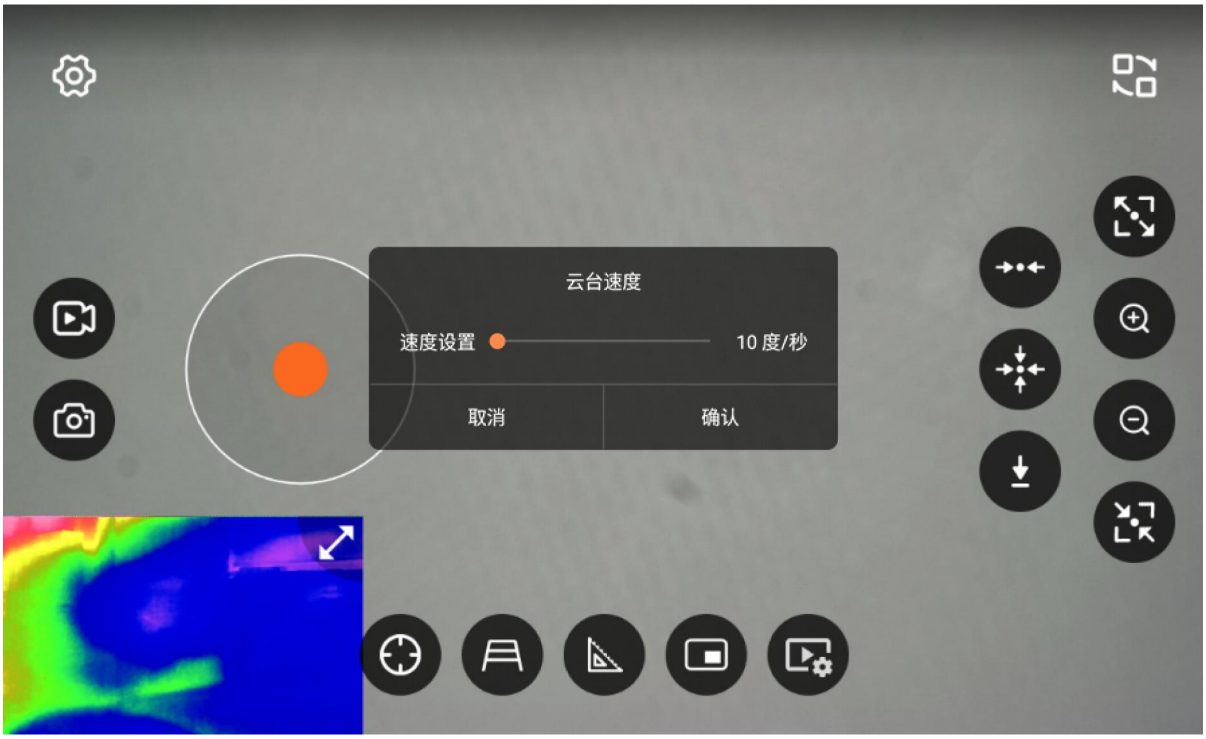
5.5.5 工作模式



可设置云台为吊装或者倒装模式，

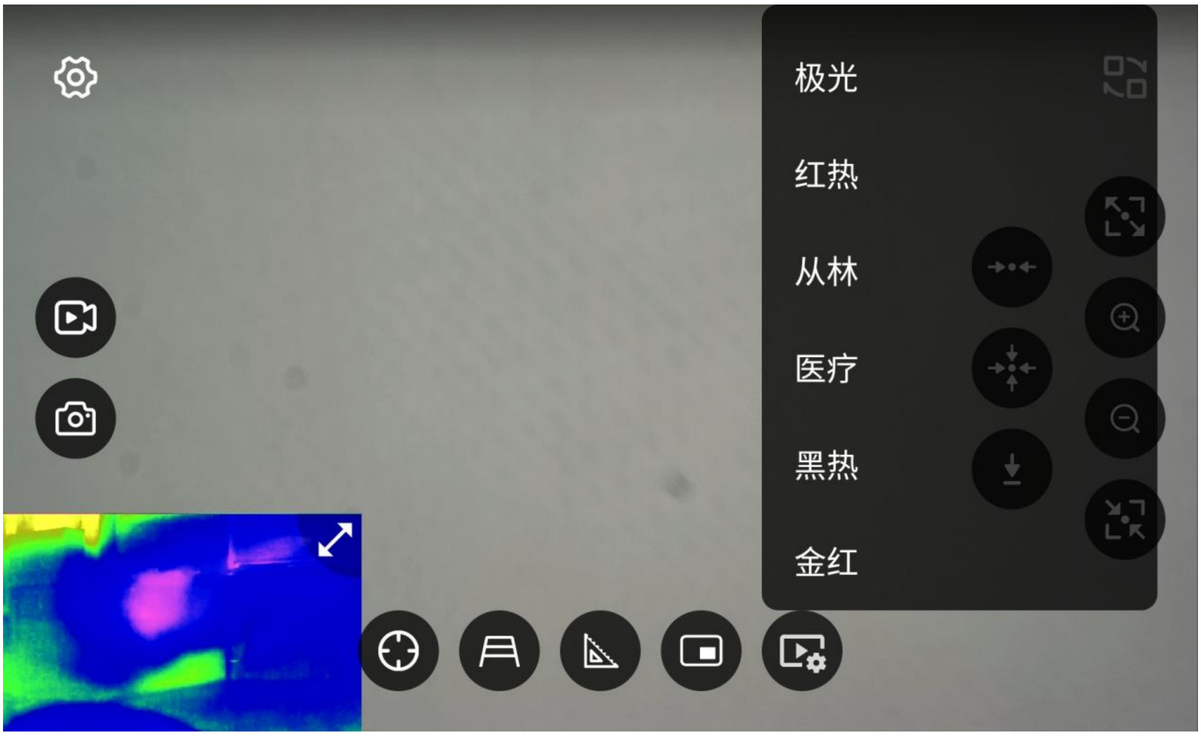
注意：设置好工作模式后，云台将会自动重启，图像自动进行翻转，必须将云台进行翻转安装为设置对应的模式放置，错误的放置方式将会导致云台电机自检不过导致损坏。

5.5.6 云台速度



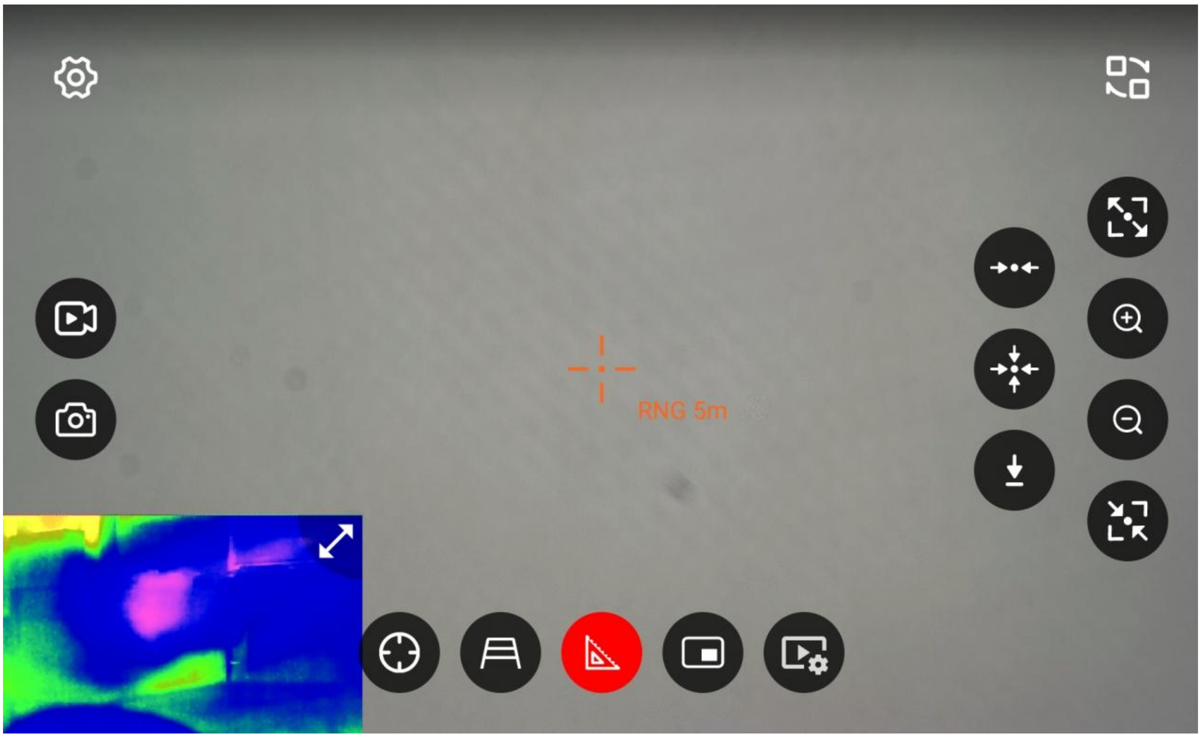
可设置云台的控制速度。

5.5.7 伪彩模式



可调热成像相机的成像效果，共有十一种可选。

5.5.8 激光测距



点击测距开关按钮可打开或关闭激光测距，有效测距范围0.1M~1.2KM

5.6 设置编码模式

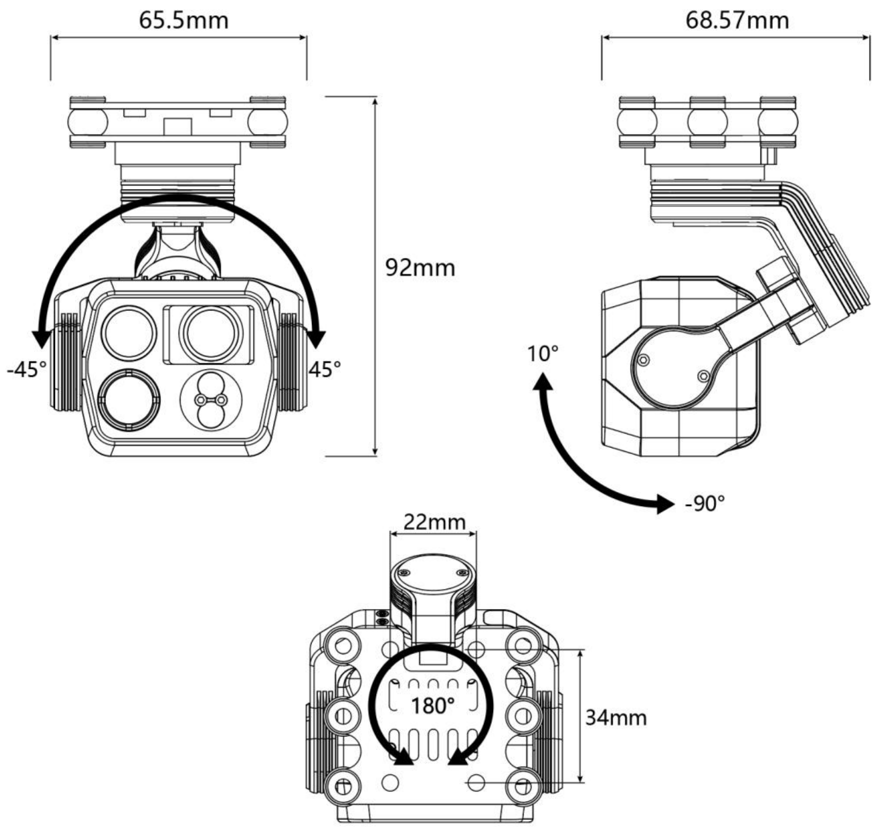
可设置画面翻转、查看相机版本号以及OSD显示。

注意：设置图像翻转后，相机将会自动重启。

5.7 固件升级

可升级云台固件和相机固件升级过程中请勿断电或者退出升级界面。（升级相机固件时C14PRO必须插TF卡）

六、云台尺寸，角度标注



七、注意事项

为保护您和他人免受伤害或保护您的设备免于损坏，请阅读以下全部信息后再使用您的设备
。

1.请勿将组件直视太阳等高强度辐射源；
2.理想使用环境温度为-10℃~60℃;
3.请勿用湿手触摸设备和线缆；
4.请勿弯折或损坏各连接线缆；
5.请勿用稀释剂擦洗您的设备；
6.请勿在未断开电源的情况下拔插其他电缆；
7.请勿接错附带的连接线缆，以免损坏设备；
8.请注意防止静电；
9. 请勿拆卸设备，如有故障请与本公司联系，由专业人员进行维修

因版本演进及客户需求变更,相应命令及控制会有所变更. 请联系云卓科技有限公司, 来获取最 新资讯及技术支持.因产品更新升级，尺寸重量等参数可能会有变化，敬请谅解。

温馨提示：使用前请仔细阅读操作说明书！

微信公众号

此说明书不一定为最新版本，最新版本说明书请移步至官网资料下载中心-资料文档获取下载：https://www.skydroid.xin/#/serviceSupport/downloadCenter

C14PRO User Manual

Small HD three-axis four-light Stabilized gimbal

V1.1

After sales service terms

1. This clause is only applicable to the products produced by skydroid, and the products sold by Sky-droid authorized dealers are also applicable to this clause.
2. As of the date of purchase, if the product of our skydroid is verified by our technicians as a quality problem caused by non-human factors within one week, skydroid shall afford the shipping fee for the repaired product. If the skydroid product is verified as a quality problem by our technicians within one week and one year after purchase, the user and skydroid shall afford the shipping fee for sending the repaired product respectively.
3. Purchase voucher and warranty card or on-line transaction record shall be provided when repairing.
4. Within 7 days from the date of purchase, skydroid products have quality problems that are not caused by human activities under normal use, and the appearance is not damaged. With the warranty card and purchase proof, the users can negotiate to replace products of the same model for free from our local dealers; When receiving the replacement product, the dealer is kindly requested to inform skydroid of the replacement at the first time.
5. skydroid will provide lifelong after-sales service for skydroid products. If the quality problem is caused by non-human factors, it will be guaranteed free of charge within one year; The user shall pay the round-trip shipping and maintenance fees for man-made damage, modification, dismantling and exceeding the one-year free warranty period from the date of purchase.
6. To ensure your rights and interests are protected and that you can be served in a timely and effective manner, please complete the warranty card and ask for the purchase voucher when purchasing Sky-droid products. Users shall provide warranty card and purchase certificate to enjoy the after-sales service terms.
7. The repaired products will be sent back to the customer within 15 working days after receiving by skydroid, and the repair report will be enclosed.
8. The above after-sales service terms are limited to skydroid products sold in Chinese Mainland.

Send the after-sales problems of Hong Kong, Macao, Taiwan and overseas customers to the email: sales01@skydroid.xin ,the specific after-sales rules depend on the situations.

Contents

Warnings and Disclaimers

The content mentioned in this document is related to your safety, legitimate rights, interests, and responsibilities. Before using this product, please read this document carefully to ensure that the product has been correctly set up. Failure to comply with and operate according to the instructions and warnings in this document may cause harm to you and people around you, and may damage this prod-uct or other surrounding items. skydroid Technology reserves the final right to interpret this document and all related documents. Updates will be made without prior notice. Please visit the official website at https://skydroidglobal.com for the latest product information.

The C14PRO gimbal has been debugged according to the camera and lens it is equipped with before leaving the factory. Do not adjust the gimbal or change its mechanical structure on your own, and do not add other external devices to the gimbal. The gimbal has a precise structure. Do not disassemble or assemble the C14PRO by yourself, otherwise it may cause abnormal operation of the gimbal camera.

For the safety of you and others, please ensure that the aircraft is in a safe state during the gimbal debugging. Do not power on and debug in an environment with flammable, explosive substances or in the presence of children. It is highly recommended that you remove the aircraft's propellers before debugging.

A) Ambient temperature: -10°C to +60°C B) Storage temperature: -25°C to +60°C C) Atmospheric pressure: 86 kPa to 106 kPa

D) The use location should not have explosive media. The surrounding media should not contain gases that corrode metals and damage insulation, nor conductive media. The presence of excessive water vapor and severe mold is not allowed.

E) The use location should be equipped with facilities to protect against rain, snow, wind, sand, and dust.

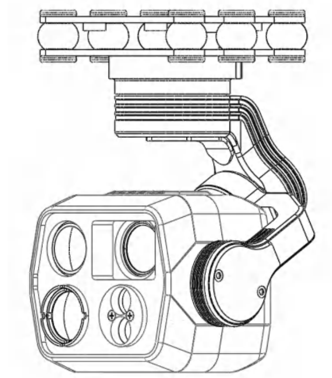
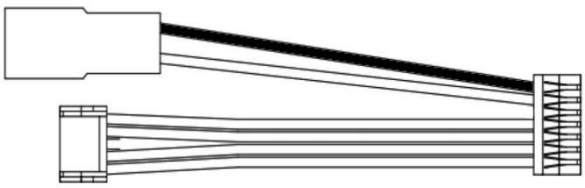
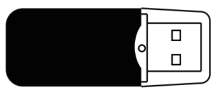
Since our skydroid has no control over specific user usage, installation, modification, and misuse, etc., our skydroid will not assume liability for any direct or indirect losses or damages caused thereby. If users install, assemble or use skydroid

Technology products, they shall bear the corresponding consequences. Our skydroid will not be held responsible for any direct or indirect losses or damages caused by the use of this product.

Ⅰ、 Product Introduction

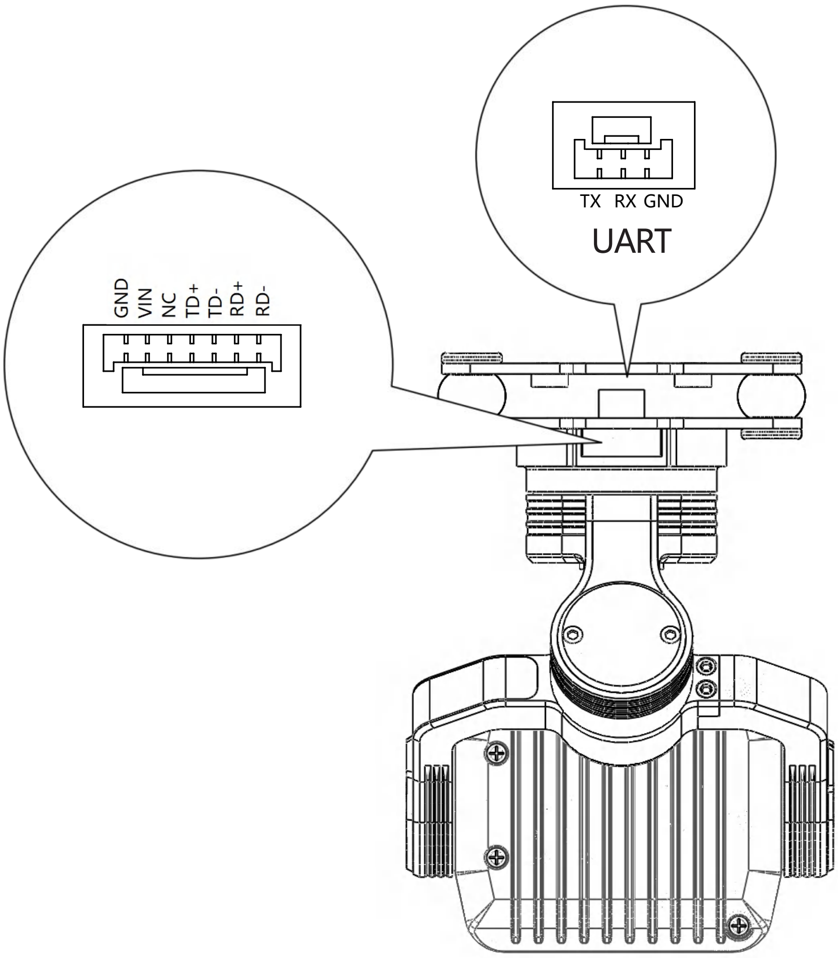
The C14PRO is a compact HD four-light mechanically stabilized gimbal. It utilizes a new generation imaging chip and it integrates distortion-free HD short-focus and long-focus cameras with an effective pixel count of 48 million, it boasts robust 4K video recording and 8K photo-taking capabilities, supporting 100x hybrid electronic zoom , which brings distant scenery into sharp focus. Equipped with a thermal imaging camera with a resolution of 640×512, it offers a wide field of view and clear images, enabling long - distance observation of various heat sources. In combination with the laser ranging module, the effective ranging distance can reach up to 1.2 kilometer. The industrial - grade three - axis stabilization structure is adopted, which significantly reduces frame jitter, keeping the image in a stable state at all times. It can be applied in fields such as fire rescue, animal protection, and security monitoring.

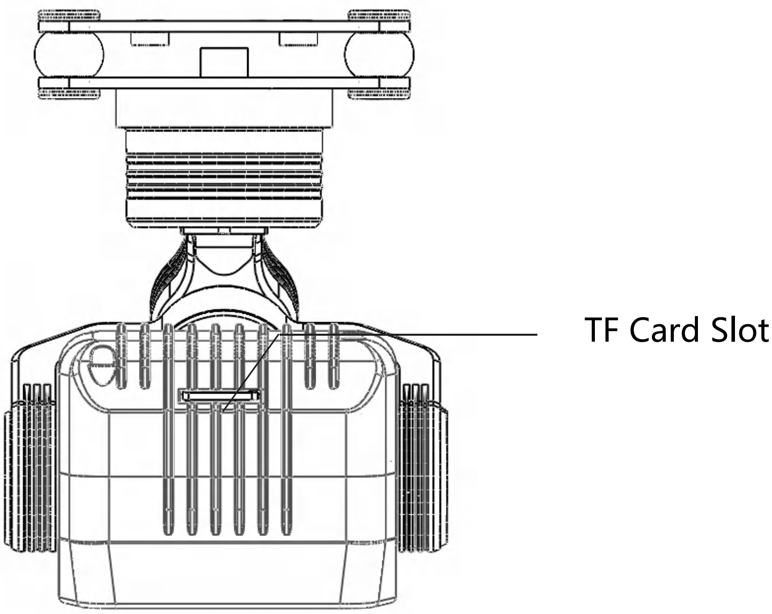
Ⅱ、 Product List

C14PRO Main Body×1	
Gimbal fixing screws (M2.5*6) ×4	
Ethernet port power cable ×1	
Memory Card × 1 (Already installed)	
Card reader×1	

Ⅲ、 Product Overview

Interface and Definition Instructions





IV、 Product Parameters

C14PRO Paramater	
Video output signal interface	Ethernet Port
Control signal input interface	Ethernet Port
Operating Voltage	15V~80V
Operating Current	15V 376mA
Operating Temperature	-10℃~+50℃
Storage Temperature	-20℃~+50℃
Weight	210g
Dimensions	65.5mm (Length) × 68.57mm (Width) × 92mm (Height)
Controllable angle range	-110° to +75° (Pitch axis); -165° to +165° (Yaw axis); -40° to +40° (Roll axis)
Stabilization Method	3-axis Mechanical Stabilization
Angular Jitter	Hovering: ±0.01° Flying： ±0.02°
Video Output	RTSP
Control Method	UDP
Ceiling-mounted/Inverted-mounted mode	supported
Global centering/One-key downward/Yaw centering	supported

C14PRO Visible Light Camera Parameters			
Lens pixel	48 million	Video resolution for card recording	3840*2160
CMOS size	1/2.0	Video storage format	H265
Zoom Ratio	100x Hybrid Electronic Zoom	Photograph resolution	8000*6000
Video transmission resolution	1280*720	Photograph storage format	JPG
Supported Memory Card Type	MicroSD Card (Max. 128GB)	Supported file systems	FAT32/exfat

C14PRO short-focus lens parameters			
Aperture	F=2.6±5%	Focal Length	5.4mm±5%
HFOV/VFOV/DFOV	61.4°/47.9°/73.8°		

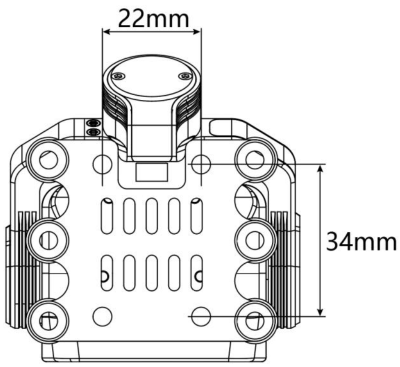
C14PRO long-focus lens parameters			
Aperture	F=2.4±10%	Focal Length	25mm±5%
HFOV/VFOV/DFOV	14.7°/11°/18.3°		

C14PRO Thermal Imaging Camera			
Resolution	640*512	Aperture	F=1.0
Focal length	11mm	Pixel pitch	10μm
Frame rate	30Hz	Response band	8~14μm
Thermal time constant	< 12ms	Field of view angle	32.84°x26.35°
Detection range	1.1km		

C14PRO Laser Ranging Parameters			
Wavelength	905nm±5nm	Detection range	0.1M~1200M
Ranging capability	0.1m to 1200m, max 1500m (night mode) Measurement range may be affected under strong sunlight or low visibility conditions	Ranging Accuracy	±0.5m (≤100m) ±0.5m+d*0.7‰ (100m<d<1200m)
Receiving Aperture	Φ6.5mm	Measurement Frequency	1.5~10Hz
False Alarm Rate	≤1%	Accuracy rate	≥98%

V、 Installation and Debugging

5.1 The schematic diagram of screw holes and their spacing is as follows:

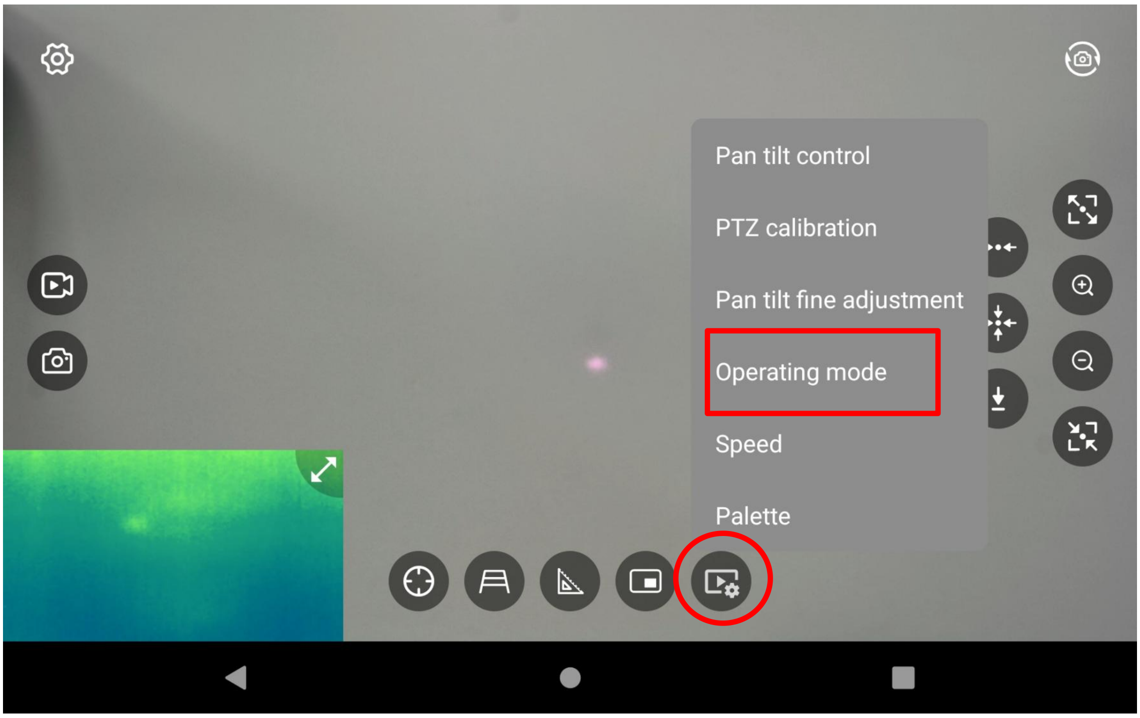


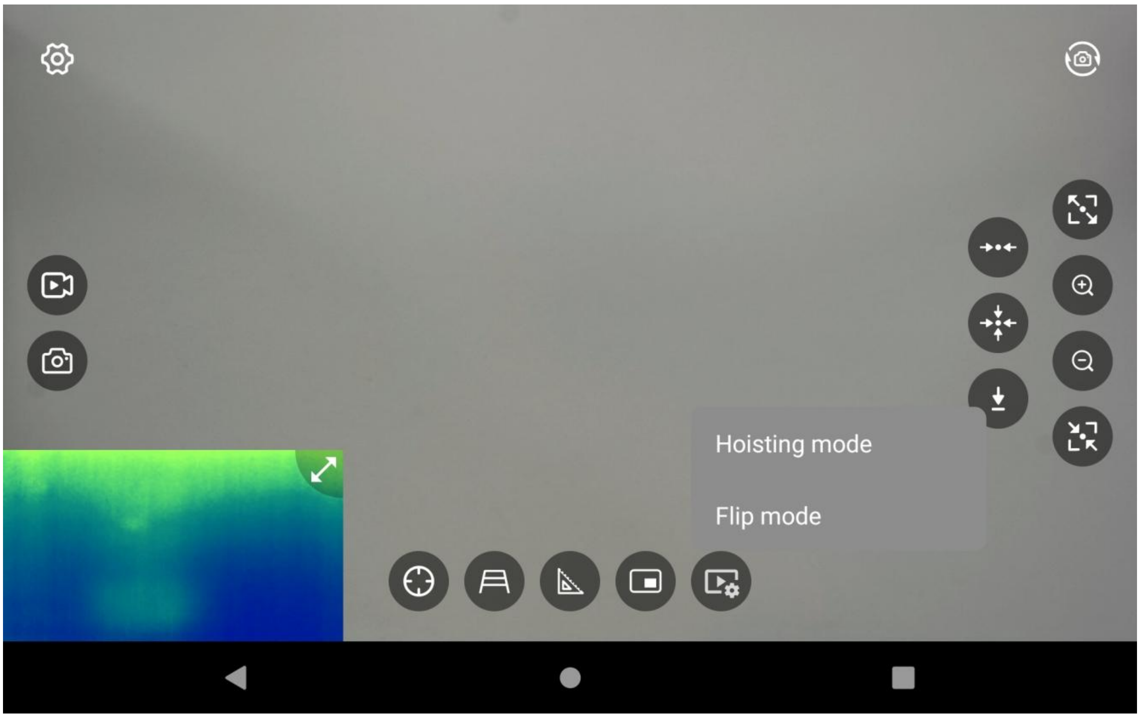
Note: The screw specifications for the pan - tilt are M2.5*6, and the quantity is 4.

Do not hang the gimbal on the aircraft when it is not in use. Long - term hanging will accelerate the deformation of the shock - absorbing balls, resulting in a decline in shock - absorbing performance and the appearance of the "jello" effect.

When installing the gimbal, the shock - absorbing plates must be kept absolutely perpendicular and parallel to each other. Incorrect installation will cause the shock - absorbing balls to deform, leading to a decrease in shock - absorbing performance and failure of self - inspection.

5.2 Gimbal Working Modes

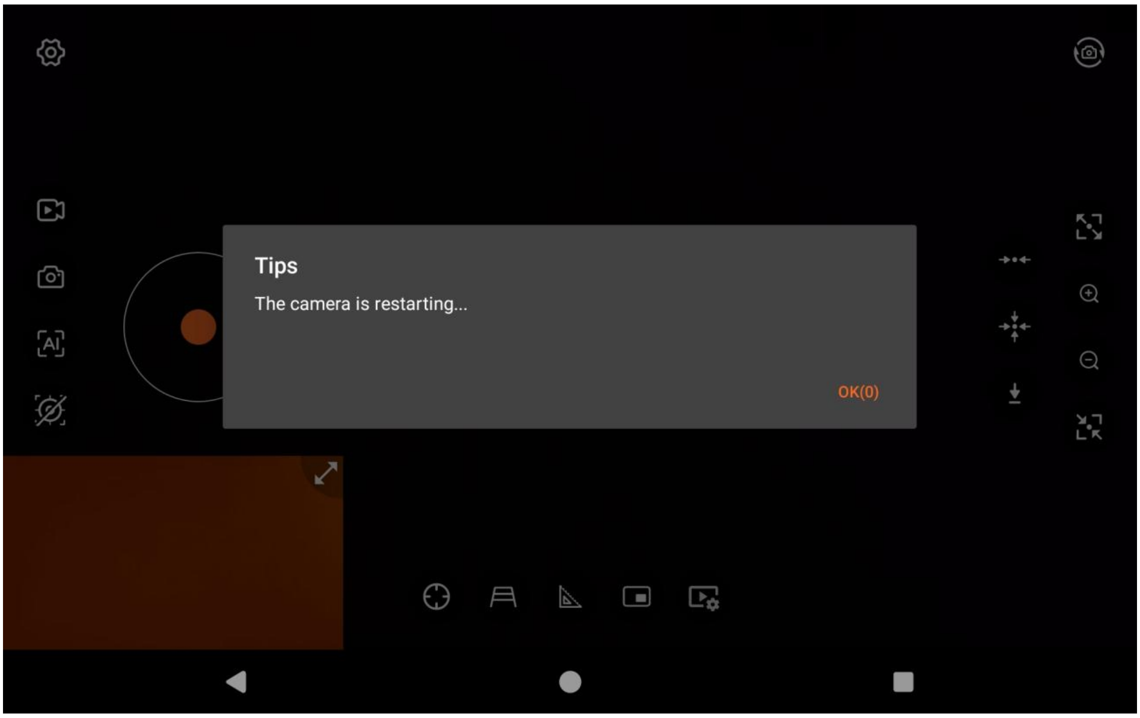




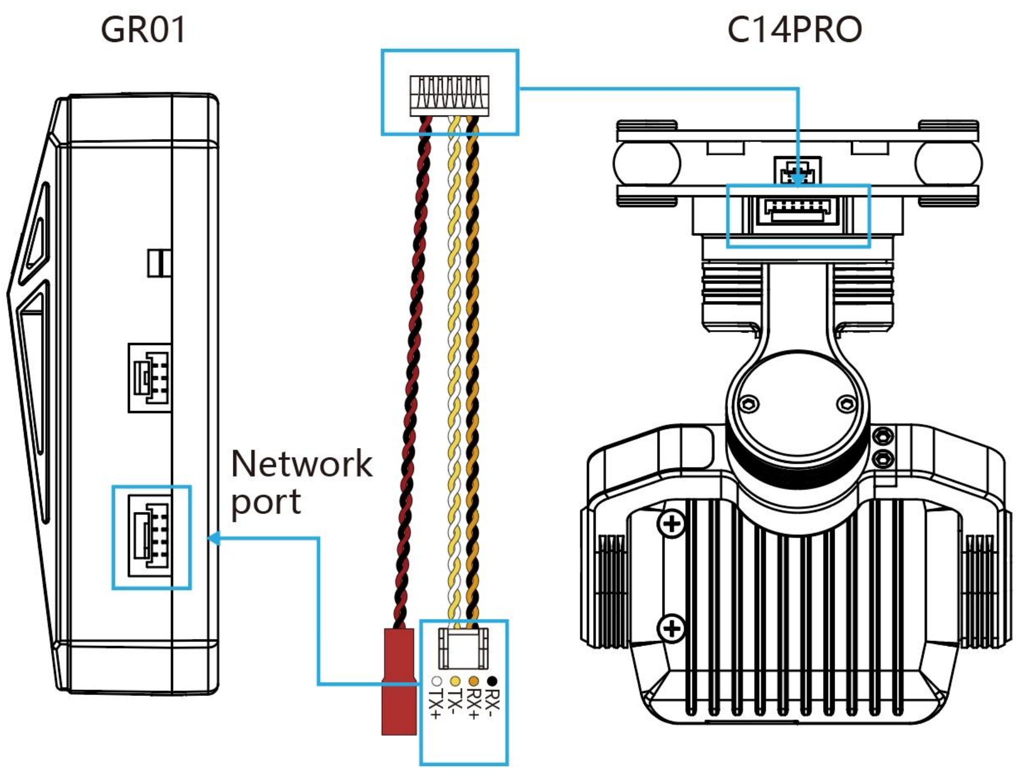
You can set the gimbal to the hoisting mode or the filp mode in the FPV settings.

Attention： The gimbal is set to hanging mode by default out of factory. DO NOT power on it when mounted upside down.

After configuring the working mode, the gimbal will restart automatically and the video image will flip accordingly. You must remount the gimbal in the orientation matching the selected mode. Improper installation will cause gimbal motor self-test failure and permanent damage.



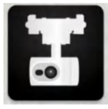
5.3 Gimbal Wiring Diagram and Explanation



power	Red JST - 2P male connector. Power supply: 15V - 80V (DC power supply or lithium - battery)	
Ethernet cable (for signal transmission)	Nerwork IP signal	RX-: Network IP signal RX+: Network IP signal TX-: Network IP signal TX+: Network IP signal
Video Transmission	Visible light video output, RTSP stream rtsp://192.168.144.108:554/stream=1	
	Thermal imaging video output, RTSP stream rtsp://192.168.144.108:555/stream=2	

5.4 Usage of C14PRO

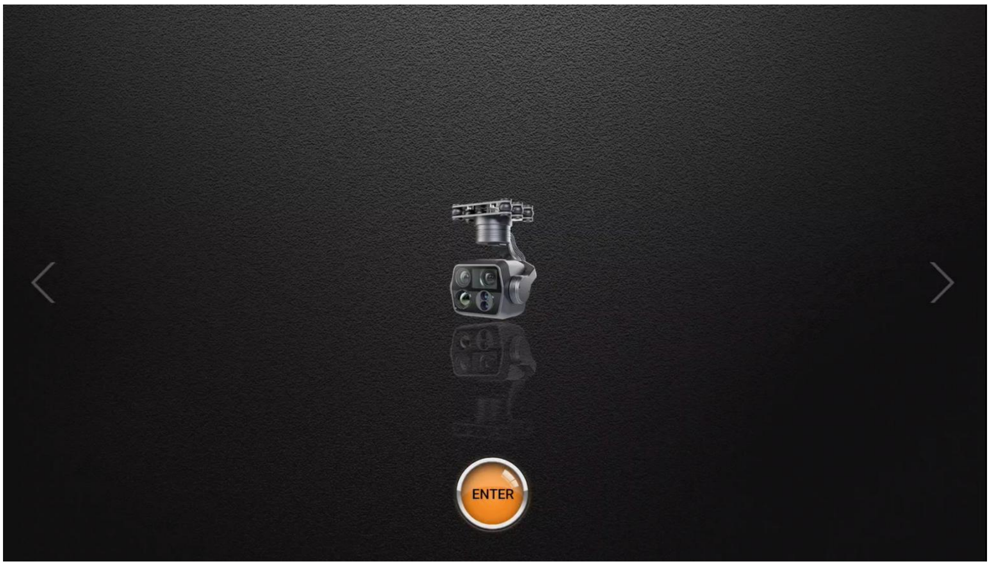
Install and open the latest version of the Gimbal FPV software on the remote control.

Software download address: http://file.skydroid.xin/skydroidCameraFPV.apk 

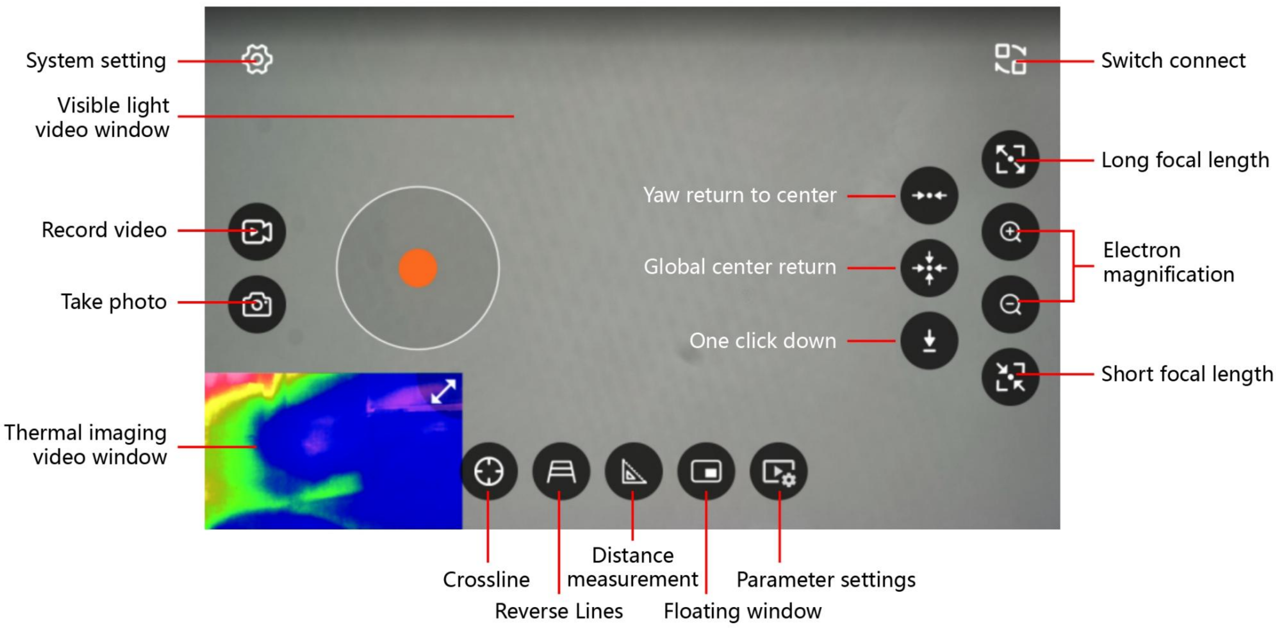
① Gimbal startup

After the gimbal is properly installed and fixed, power it on. The gimbal will start up. Please wait until the startup is complete.

② Open the Gimbal FPV software, select the C14PRO, and click to enter.

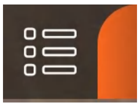
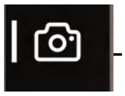


③ Introduction to the Home Page of the Gimbal FPV

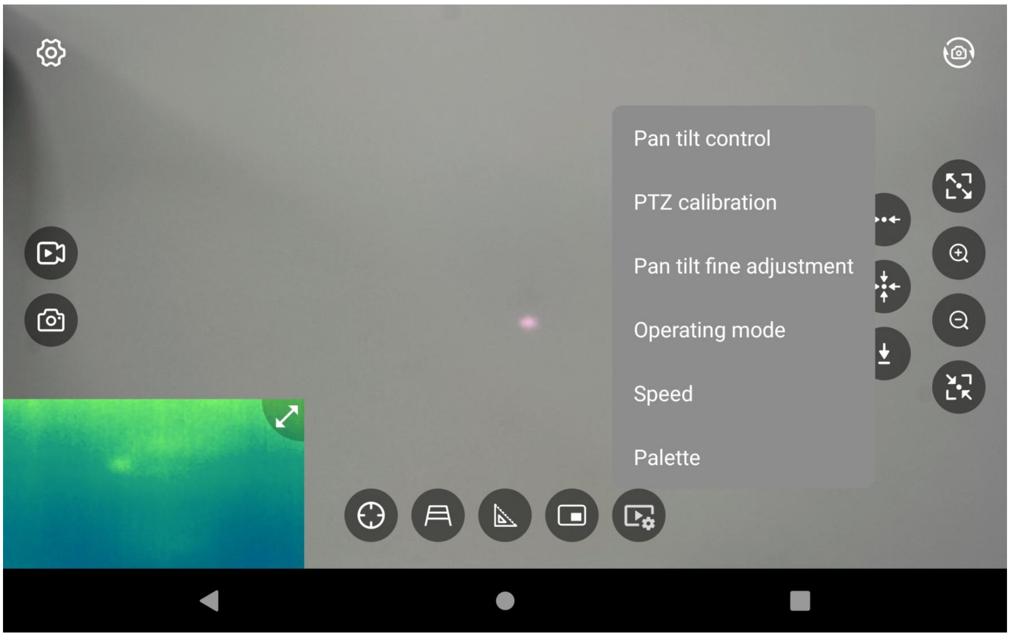


④ FLY GCS Screen Settings

Skydroid FLY Download Link: http://file.skydroid.xin/skydroidFly.apk

On the top-left corner of the APP’ s homepage  ->  ->Tap to open the left window and set it to C14/C14PRO.

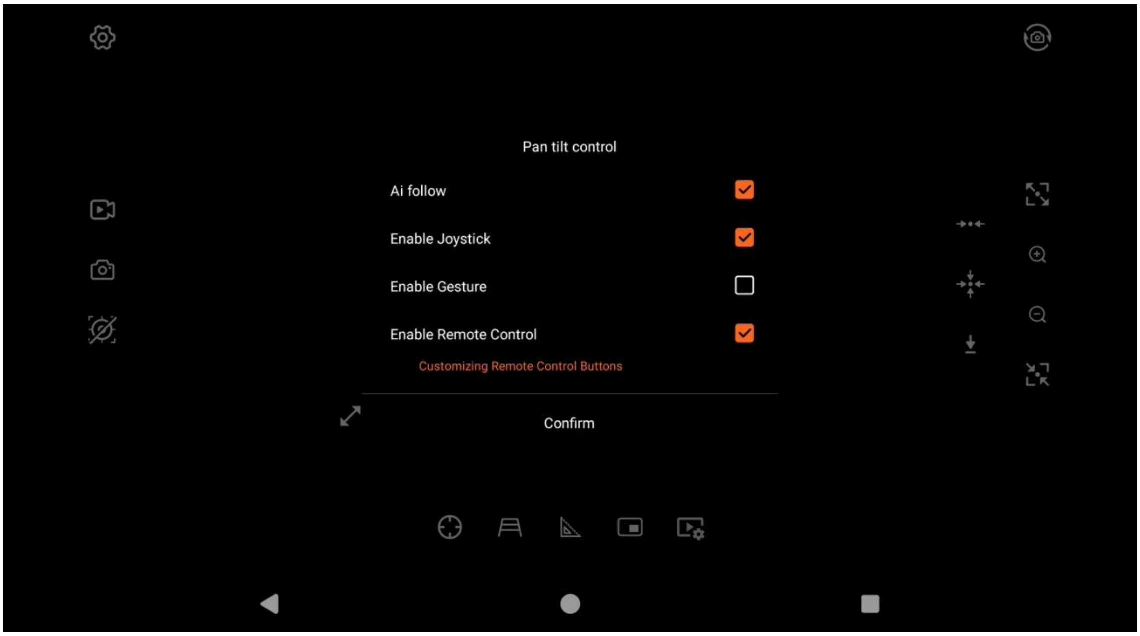
5.5 Gimbal Parameter Settings



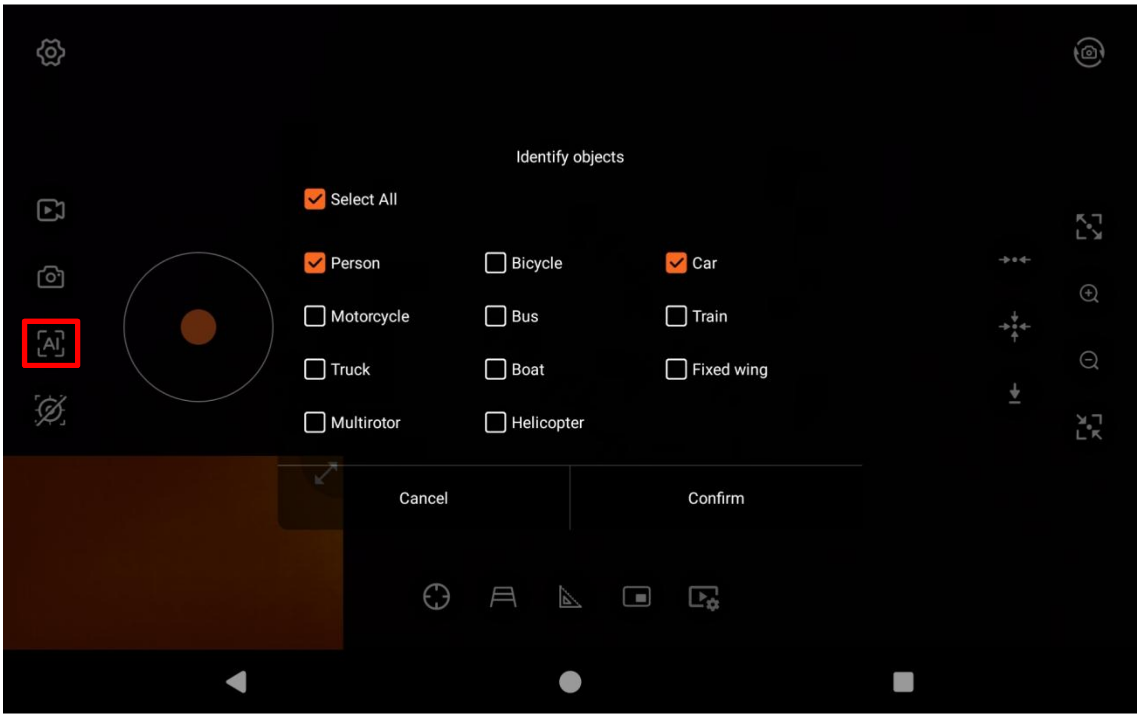
Gimbal control, calibration, fine-tuning, operating modes, speed, and color plaette can be configured in the parameter setting

5.5.1 Gimbal Control

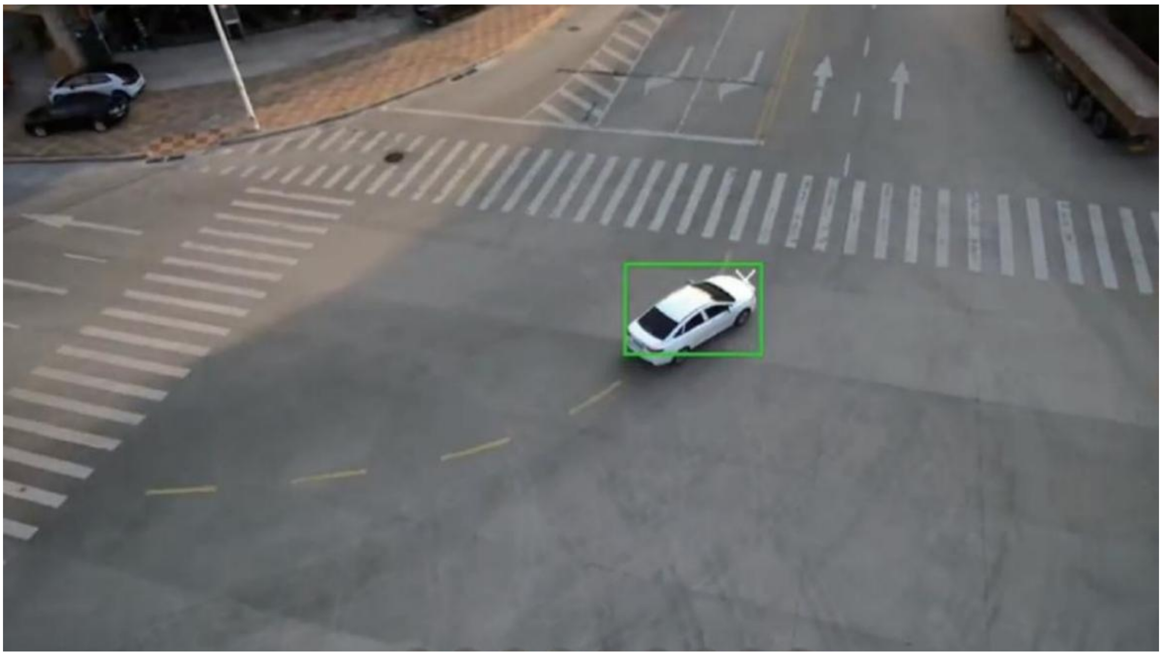
Click Parameter Settings -> Pan-tilt Control, there are target following and three control methods to choose from, and multiple methods can be selected simultaneously. (Target following cannot be used simultaneously with gesture control.)



① Gimbal AI Object Tracking, tap the AI icon to select single, multiple or all targets for automatic identification and tracking.

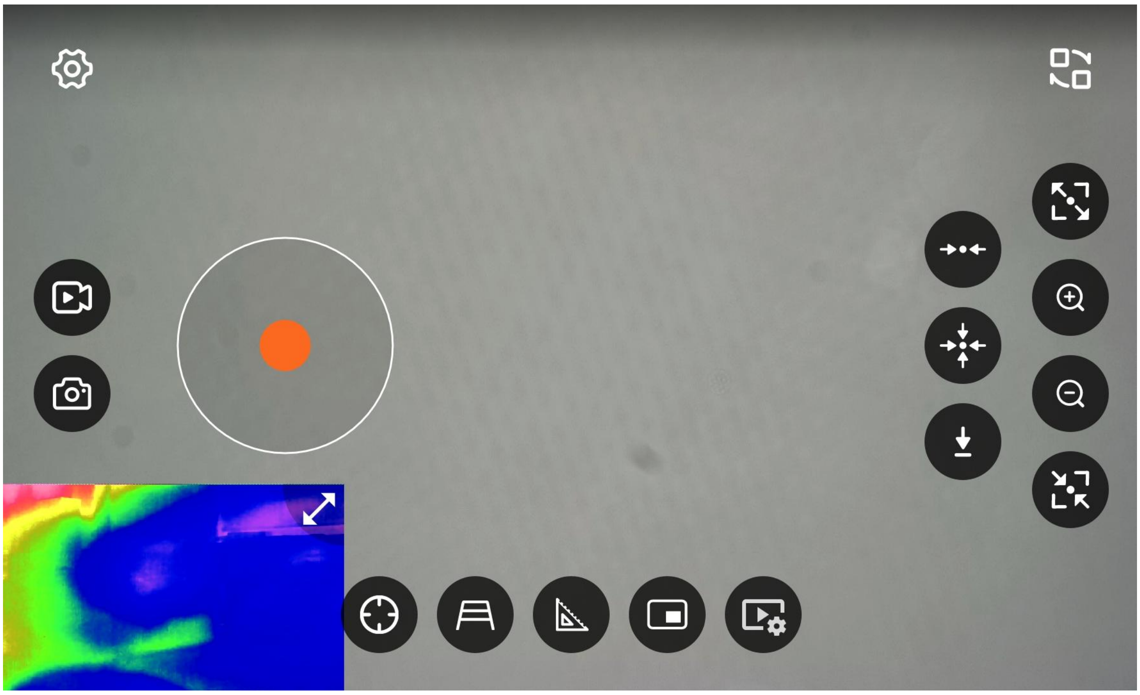


C14PRO will identify objects automatically (marked with green frames for detected targets).

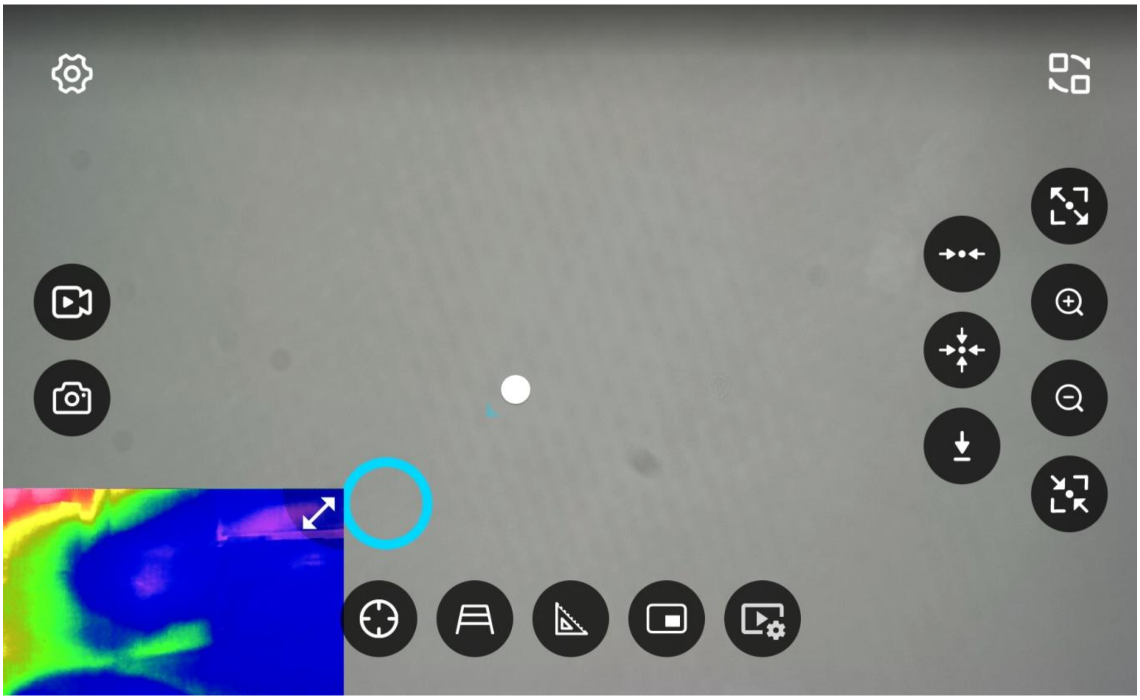


Tap the target inside the frame to lock it (the frame will turn white). The C14PRO will then track the target autonomously.

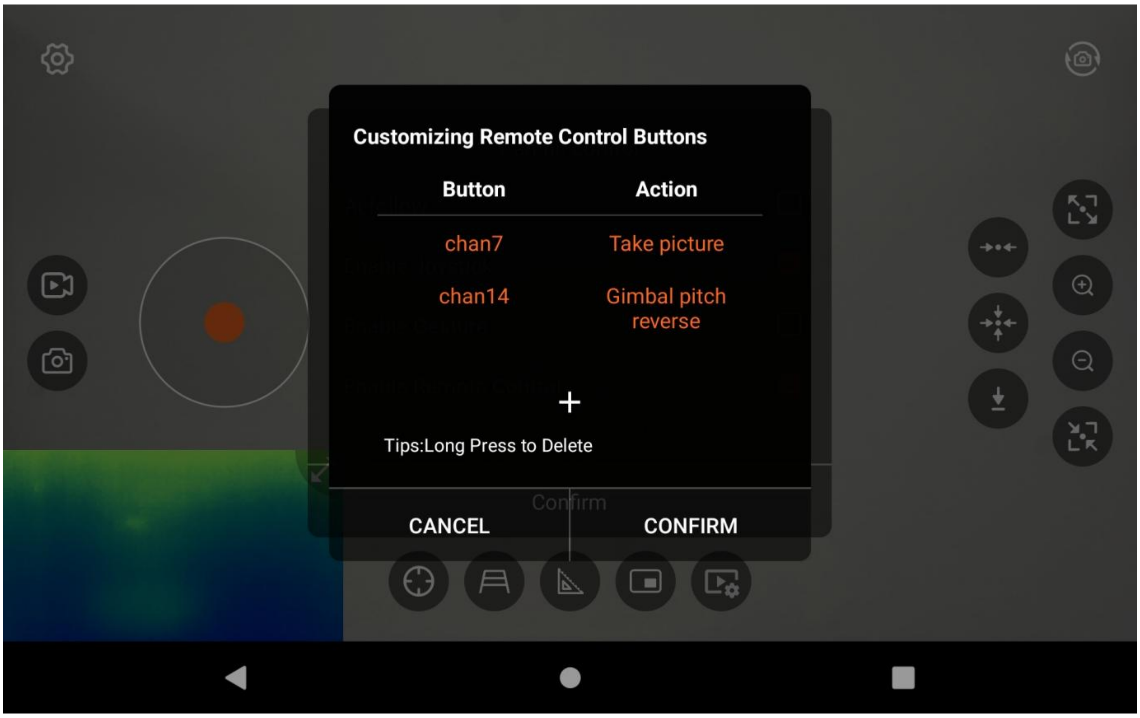
② Virtual joystick control: You can use the virtual joystick to control the pan and tilt of the gimbal. It also supports functions such as one - key return to the center.



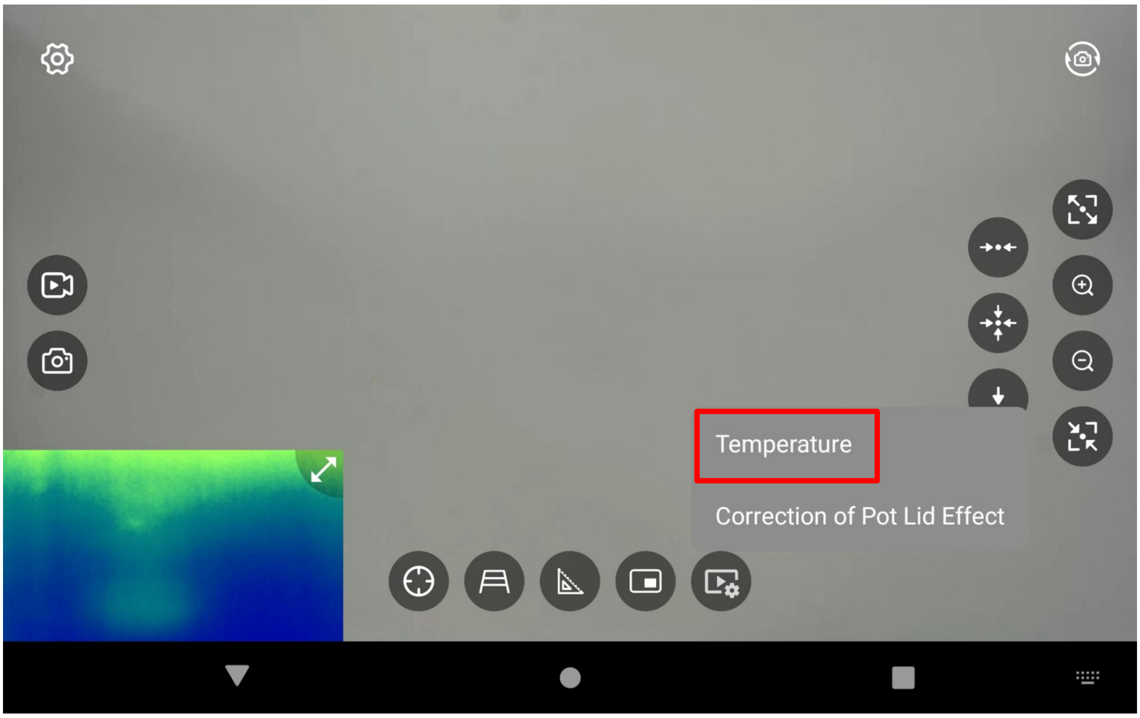
③ Gesture control: Control the pan and tilt of the gimbal by swiping the screen.



④ Remote control channel control: Control the pan and tilt of the gimbal, as well as functions like taking photos and recording videos through customizing the remote control channels. (The remote control channels can be checked through the Remote Control Assistant -> Rudder Volume View.)

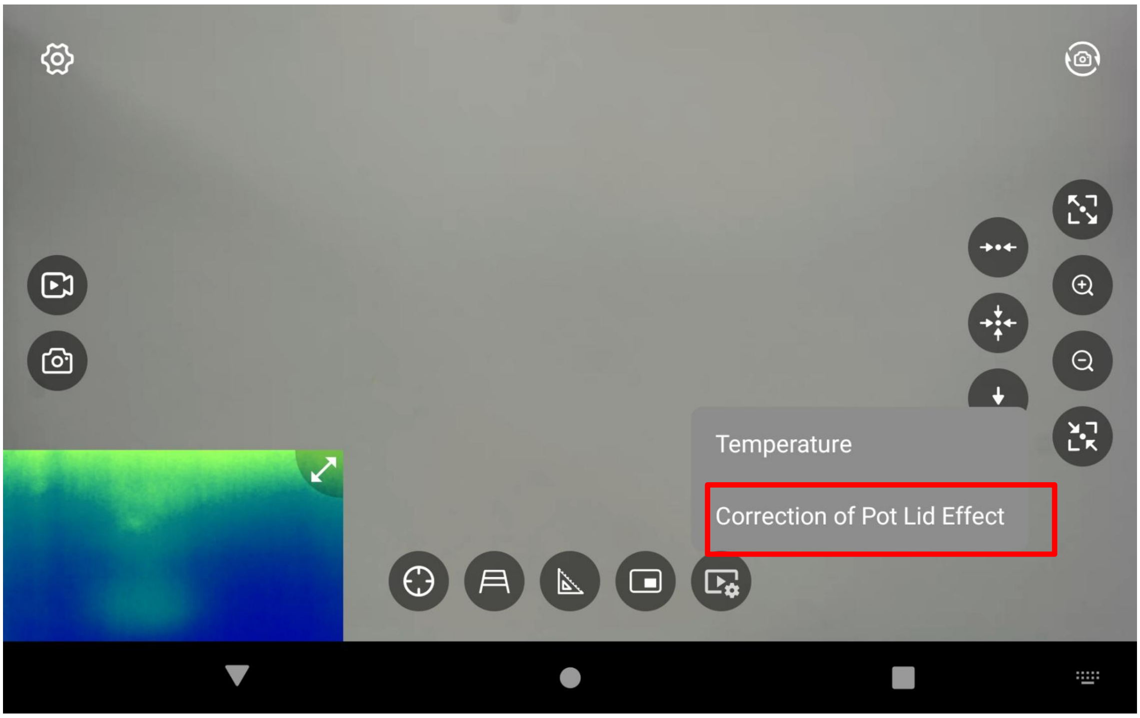


5.5.2 Gimbal Calibration



Temperature Calibration This function calibrates the operating temperature of the gimbal, preventing malfunctions caused by excessive temperature deviation between the ambient environment and the IMU. Temperature calibration is completed at the factory. Do not perform re-calibration unless necessary. Please contact Skydroid technical support to obtain the calibration password before any calibration.

5.5.3 Correction of Pot Lid Effect



Note： The pot lid effect has been calibrated at the factory. Due to the special calibration method, do not calibrate it yourself unless necessary. Before calibration, please contact the relevant technicians from skydroid for confirmation.

Calibration method:

① Power on the C13 and let it preheat for more than five minutes.

② Select a panel with a flat surface, uniform temperature, even frosting, and low reflectivity.

③ Move the C13 close to and make contact with the target surface. It is optimal when the imaging of the target surface covers the entire field of view without any stray light entering.

④ Click "Calibrate the Lid Effect", and the calibration is successful when the screen turns into a uniform gray.

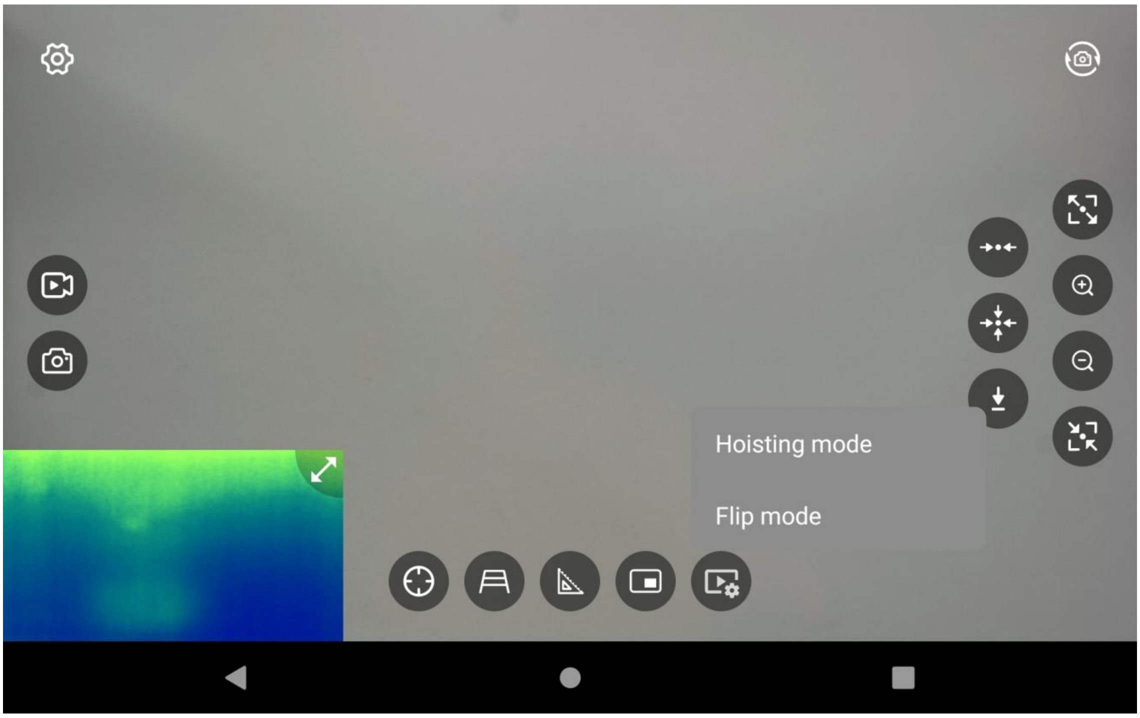
Schematic diagram of the calibration target surface for the lid - like effect

Schematic diagram of calibrating the lid effect

Gimbal Fine-Tuning: Fine-tune the gimbal’ s horizontal and pitch axes.

Note: The gimbal is factory-calibrated. Do not perform this operation unless gimbal malfunctions occur.

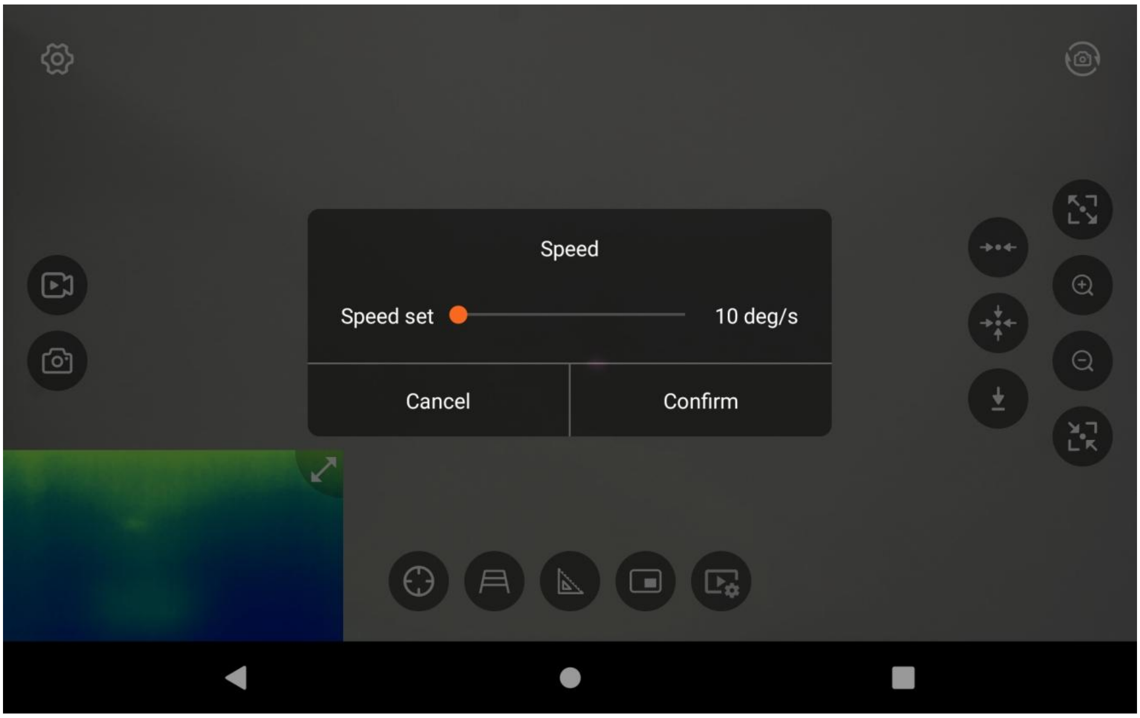
5.5.5 Operating Modes



The gimbal can be set to either top-mounted or inverted-mounted mode.

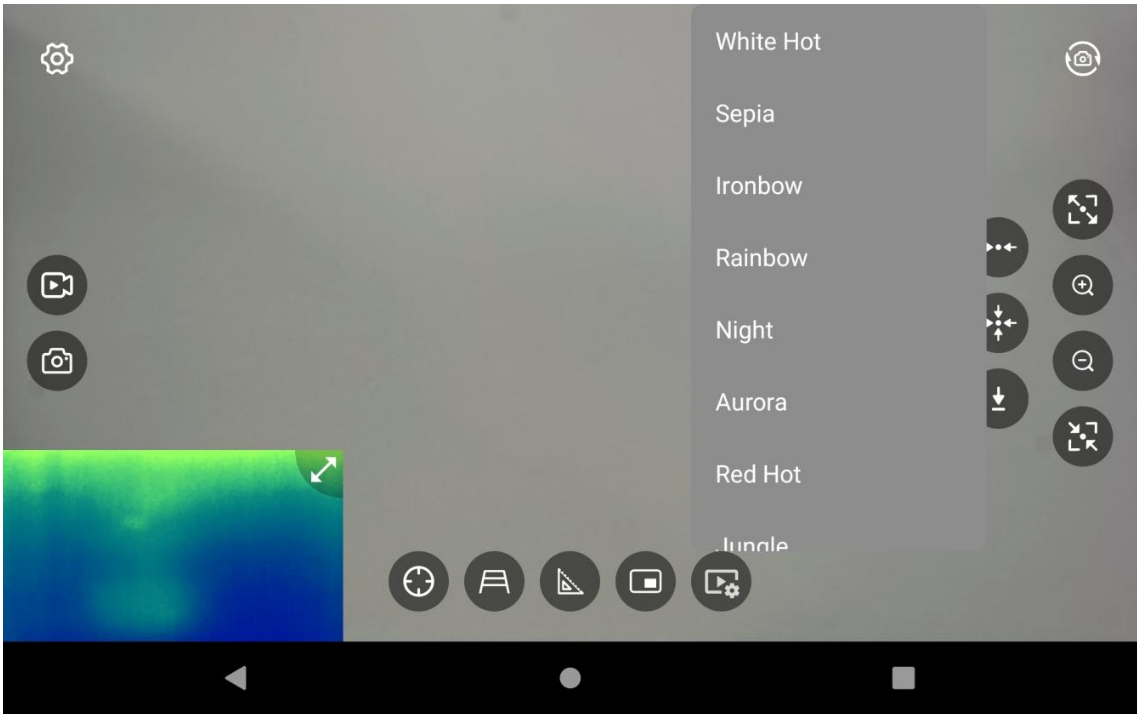
Note : After setting the working mode, the gimbal will automatically restart and the image will automatically flip. The gimbal must be installed in the flipped position to match the set mode. Incorrect placement will cause the gimbal motor to fail the self-test and result in damage.

5.5.6 Gimbal Speed



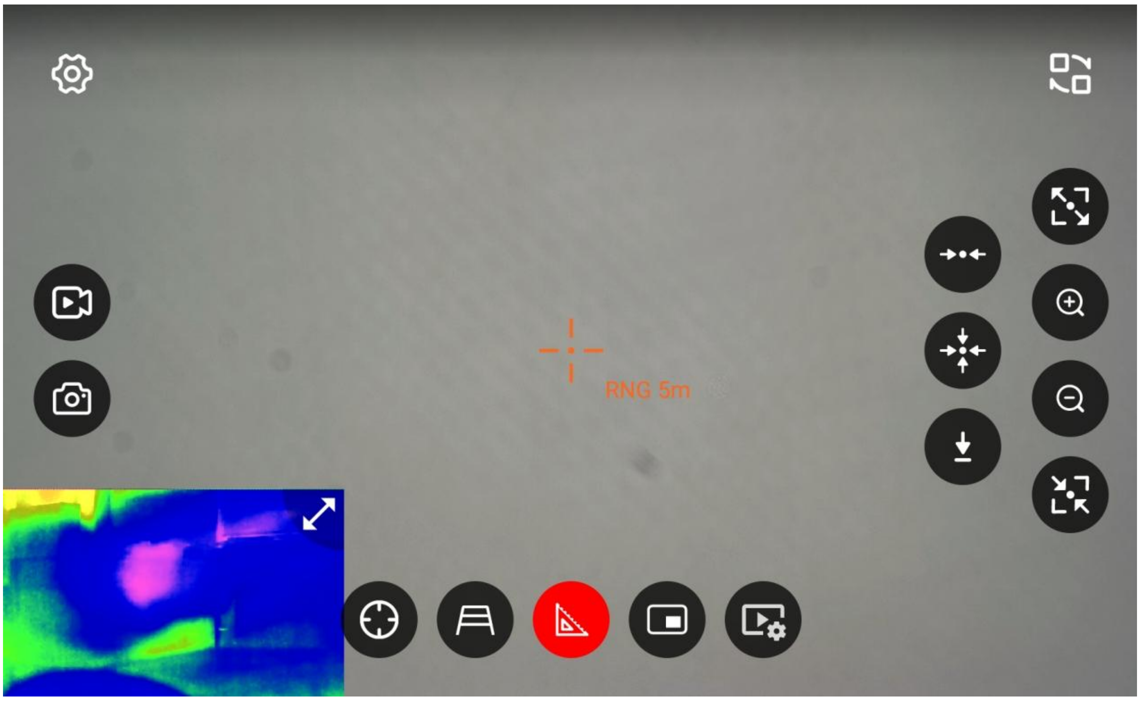
The control speed of the gimbal has two modes: constant speed mode and variable speed mode.

5.5.7 Color Palatte



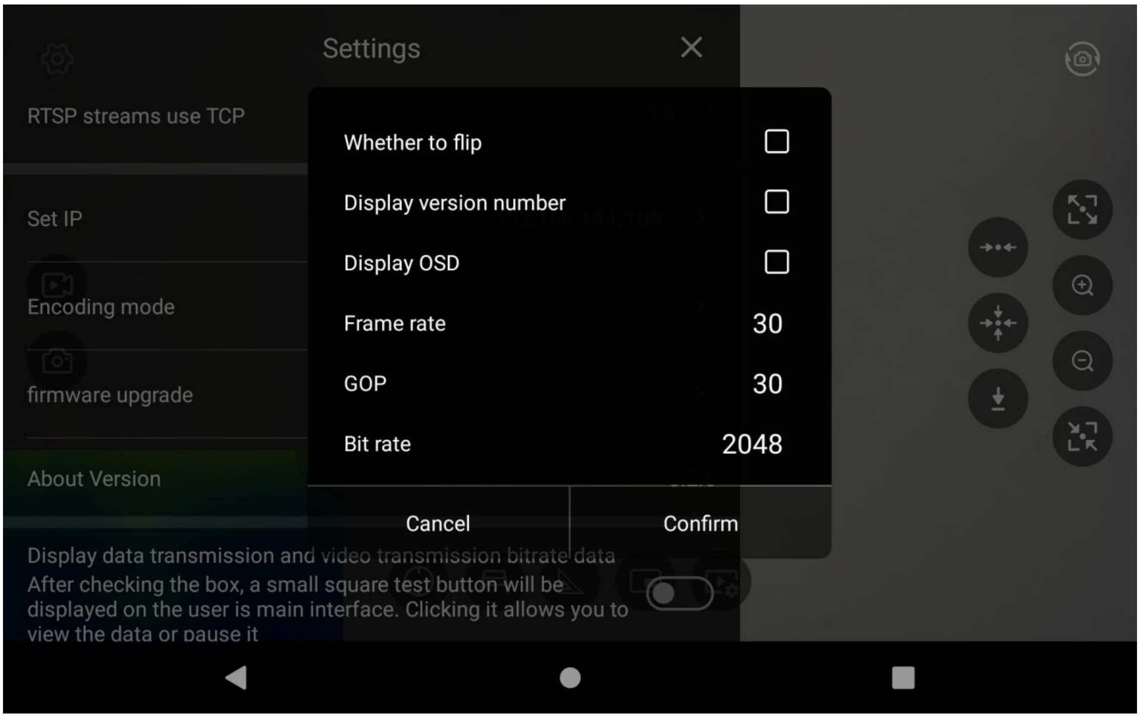
It is possible to adjust the imaging effect of the thermal imaging camera, with a total of eleven options available.

5.5.8 Laser Ranging



Press the ranging switch button to turn laser ranging on or off.The effective ranging range is 0.1m to 1.2 km.

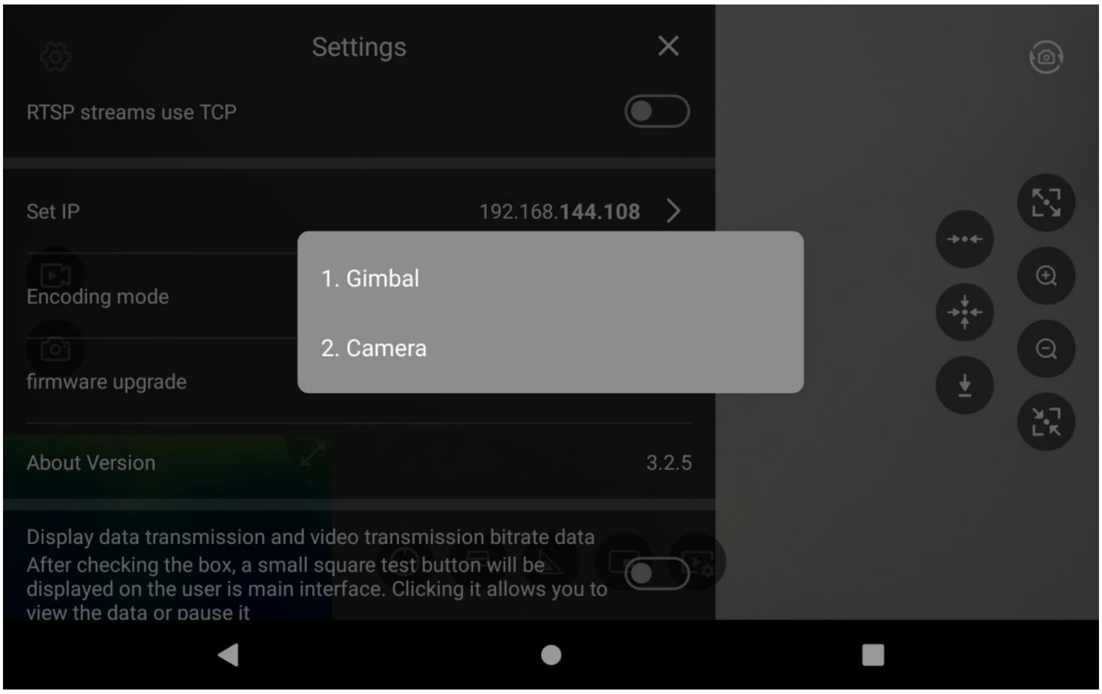
5.6 Set Encoding Mode



You can se screen flipping, view the camera version number, and set Display OSD

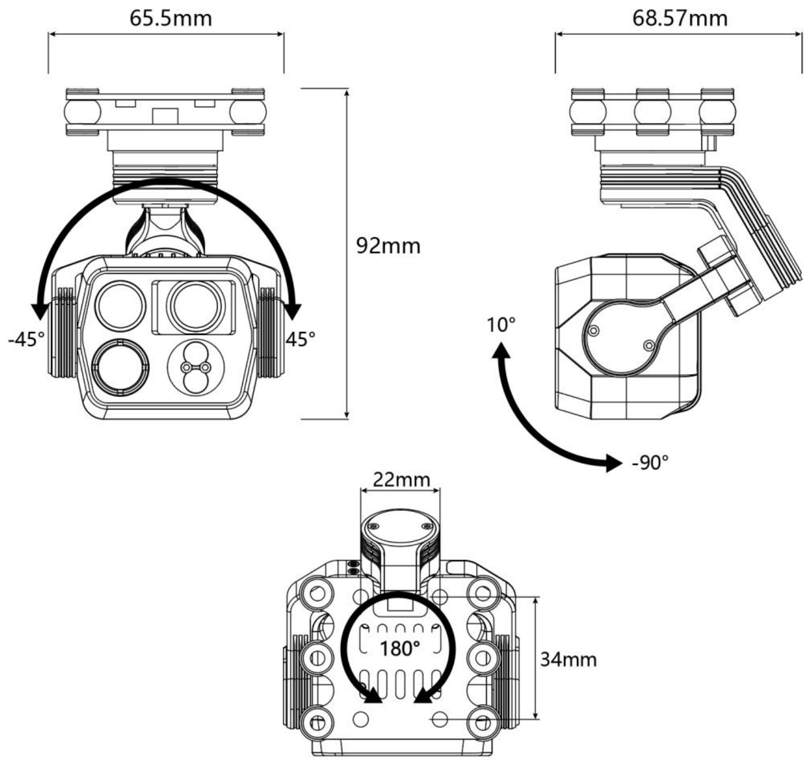
Notice: The camera will restart automatically after image flip settings are applied.

5.7 Firmware Upgrade



You can upgrade the gimbal firmware and camera firmware. Do not cut off the power or exit the upgrade interface during the upgrade process. (When upgrading the camera firmware, a TF card must be inserted into the C14PRO.)

VI、 Gimbal Dimensions and Angel Markin



As the product version evolves and customer requirements, the corresponding commands and controls are subject to adjustment. Please contact skydroid Technology Co., Ltd. to obtain the latest information and technical support. Please note that parameters such as dimensions and weight may change due to product upgrades and updates. We apologize for any inconvenience caused

Ⅶ、 Precautions

To protect you, others from harm and safeguard your device from damage, please read all the follow-ing information before using your device.

1. Do not direct components to high - intensity radiation sources such as the sun;
2. The ideal operating ambient temperature ranges from 10℃ to 60℃;
3. Do not touch the device and cables with wet hands;
4. Do not bend or damage the connecting wires;
5. Do not clean your device with thinner;
6. Do not unplug or plug in other cables without disconnecting the power supply;
7. Do not misconnect the attached cables to avoid damaging the device;
8. Be careful to prevent static electricity;
9. Do not disassemble the device. If there is a malfunction, please contact our skydroid and let professionals carry out the repair.

Due to version evolution and changes in customer requirements, relevant commands and controls may be subject to alteration. Please contact skydroid Technology Co., Ltd. to obtain the latest information and technical support. As the product is updated and upgraded, parameters such as dimensions and weight may change. Your understanding is highly appreciated.

Warm reminder: Please read the operation manual carefully before use!